Diskretni krmilni sistemi

Primož Podržaj

Fakulteta za strojništvo, Univerze v Ljubljani

5. april 2022

1. Fourier-jeva transformacija
   - Fourier-jeva vrsta
   - Fourier-jev integral
   - Lastnosti Fourierove transformacije
   - Fourier-jeve slike nekaterih pomembnejših funkcij
   - Frekvenčna karakteristika
   - Frekvenčni diagrami

Fourier-jeva vrsta

Če funkcija zadošča Dirichletovim pogojem, velja: $f(t) = A_0 + A_1\cos(\Omega t) + A_2\cos(2\Omega t) + \cdots + B_1\sin(\Omega t) + B_2\sin(2\Omega t) + \ldots$

Koeficiente izračunamo na osnovi izrazov:

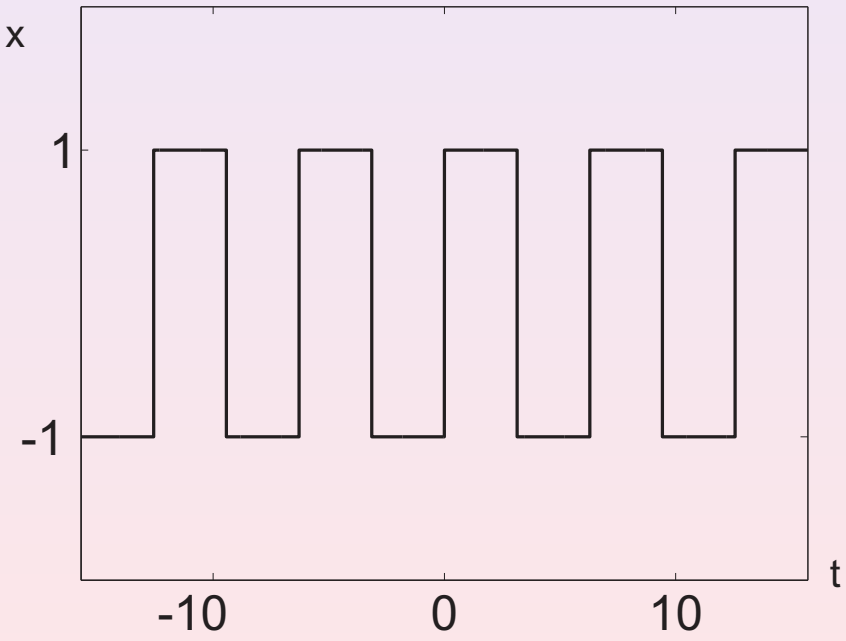
$$A_0 = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)dt$$

$$A_i = \frac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\cos(i\Omega t)dt; \quad i = 1, 2, \ldots$$

$$B_i = \frac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\sin(i\Omega t)dt; \quad i = 1, 2, \ldots$$

Primer 1-1

Za signal $x(t)$ prikazan na sliki izračunaj koeficiente v Fourierovi vrsti.



Primer 1-2

Ker je $x(t)$ liha, velja:

$$A_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} x(t)dt = 0$$

Primer 1-3

Ker je funkcija $\cos(i\Omega t)$ soda, je njen produkt z liho funkcijo $x(t)$ lih. Velja torej:

$$A_i = \frac{2}{2\pi}\int_{-\pi}^{\pi} x(t)\cos(it)dt = 0$$

Primer 1-4

Koeficienti pri sodih vrednostih $i = 2k$ so enaki:

$$B_i = \frac{2}{2\pi}\int_{-\pi}^{\pi} x(t)\sin(2kt)dt =$$

$$\frac{2}{2\pi}\left(\int_{-\pi}^{0} x(t)\sin(2kt)dt + \int_{0}^{\pi} x(t)\sin(2kt)dt\right) =$$

$$\frac{1}{2\pi}\left(\int_{-2\pi}^{0}(-1)\sin(ku)du + \int_{0}^{2\pi} 1\sin(ku)du\right) = 0$$

Primer 1-5

Koeficienti pri lihih vrednostih $i = 2k + 1$ so enaki:

$$B_i = \frac{2}{2\pi}\int_{-\pi}^{\pi} x(t)\sin((2k+1)t)dt =$$

$$\frac{2}{2\pi}\left(\int_{-\pi}^{0} x(t)\sin((2k+1)t)dt + \int_{0}^{\pi} x(t)\sin((2k+1)t)dt\right)$$

Z uvedbo nove spremenljivke $u = (2k+1)t$ dobimo:

$$B_i =$$

$$\frac{2}{2\pi}\frac{1}{2k+1}\left(\int_{-(2k+1)\pi}^{0}(-1)\sin(u)du + \int_{0}^{(2k+1)\pi} 1\sin(u)du\right) =$$

$$\frac{2}{2\pi}\frac{1}{2k+1}\left([cos(u)]_{-(2k+1)\pi}^{0} + [-cos(u)]_{0}^{(2k+1)\pi}\right) = \frac{4}{\pi}\frac{1}{2k+1} =$$

$$\frac{4}{\pi}\frac{1}{\ldots}$$

Primer 1-6

Signal $x(t)$ lahko torej aproksimiramo z naslednjo vrsto:

$$x(t) \approx \frac{4}{\pi}\sin(t) + \frac{4}{\pi}\frac{\sin(3t)}{3} + \frac{4}{\pi}\frac{\sin(5t)}{5} + \dots$$

Primer 1 - Rezultati

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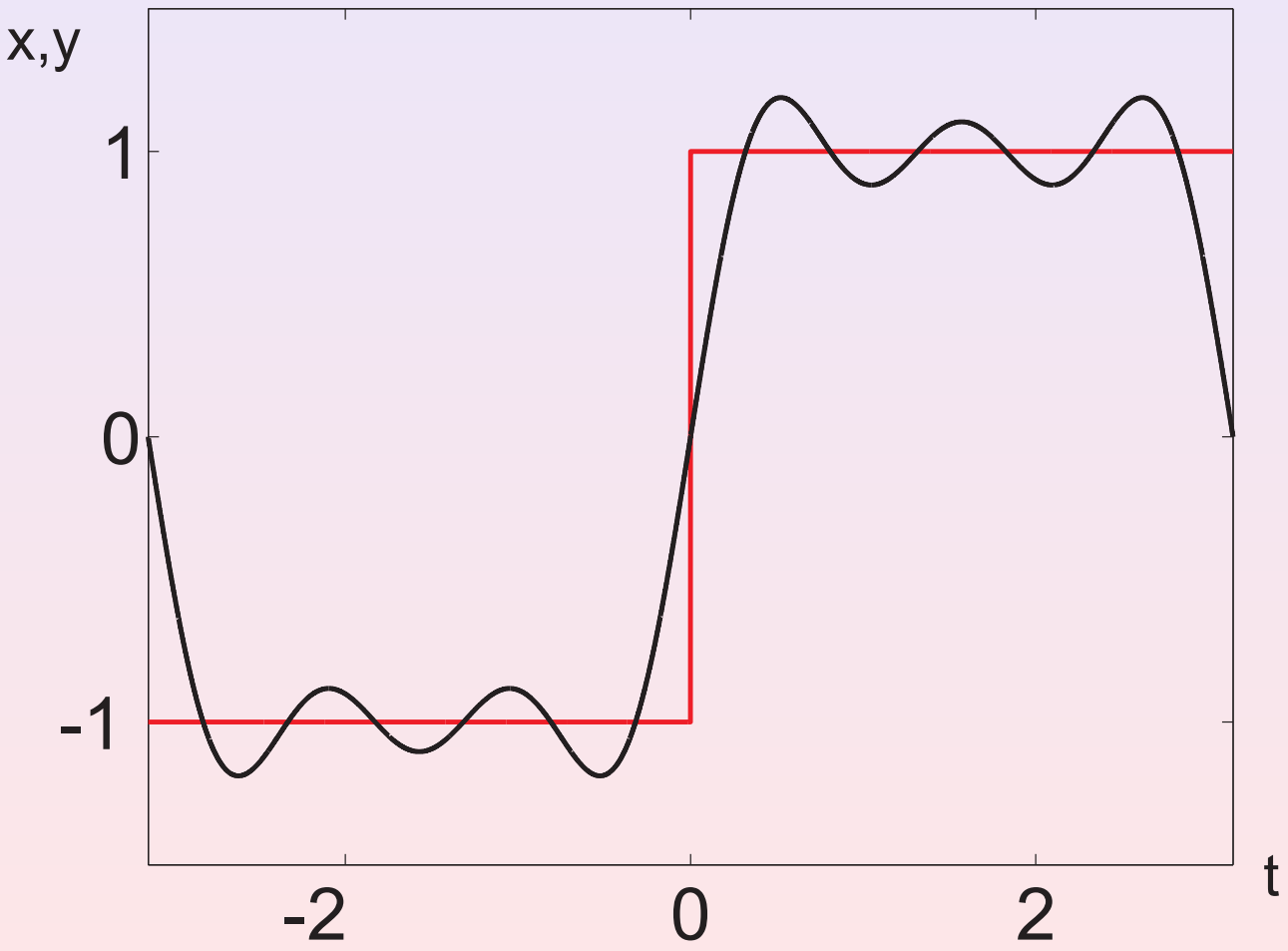

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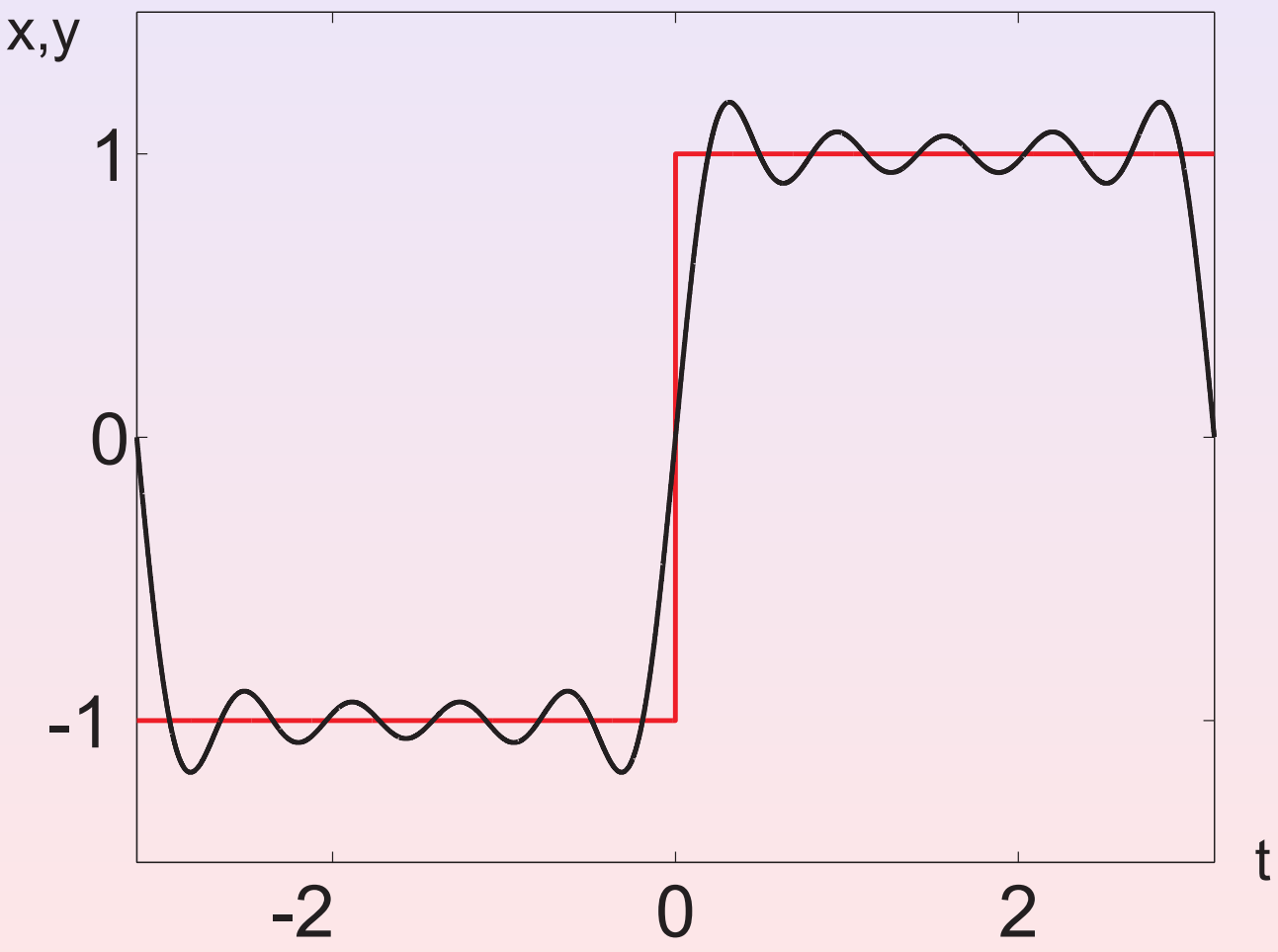

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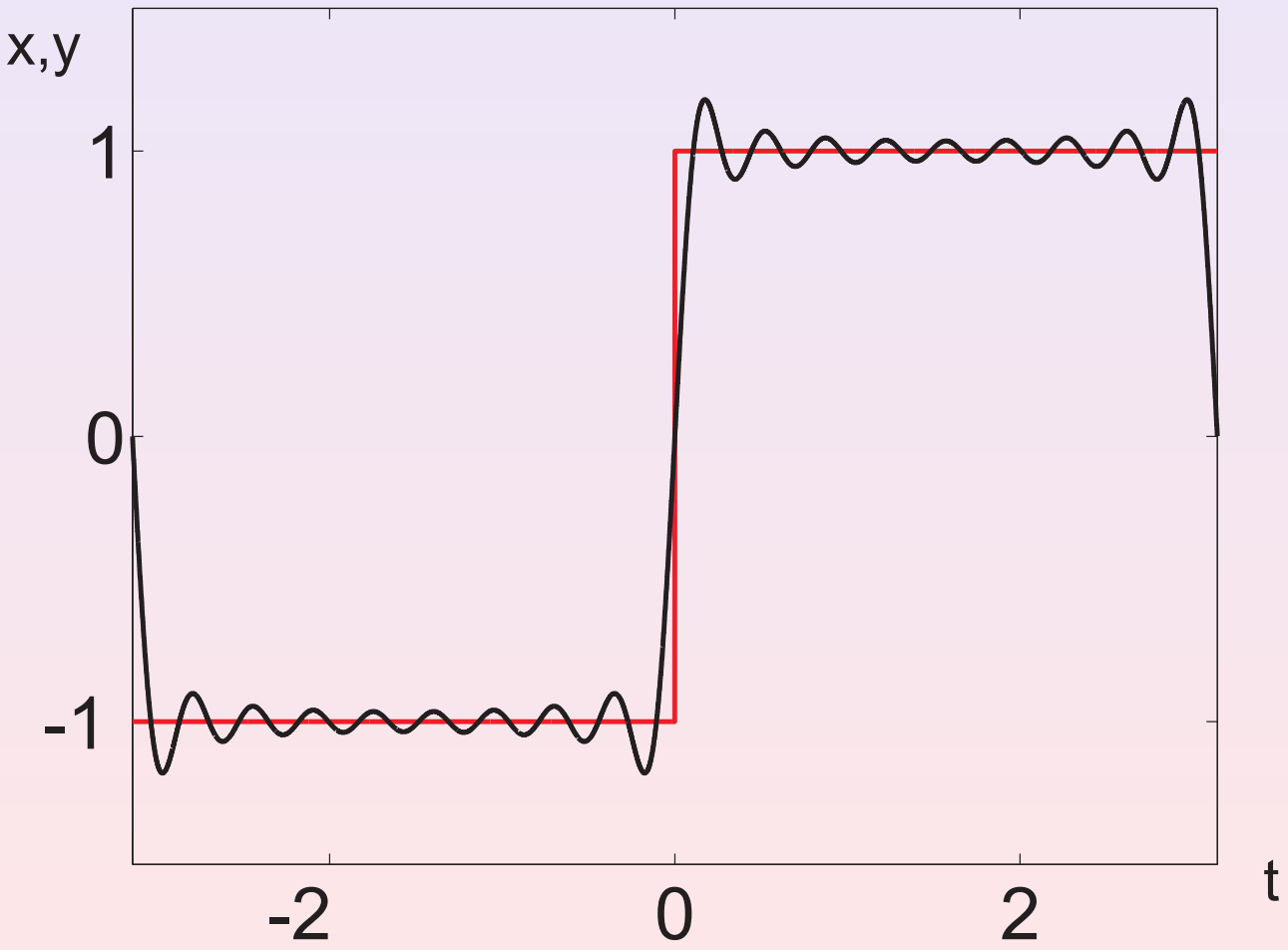

Neskončna vsota z eksponentnimi členi

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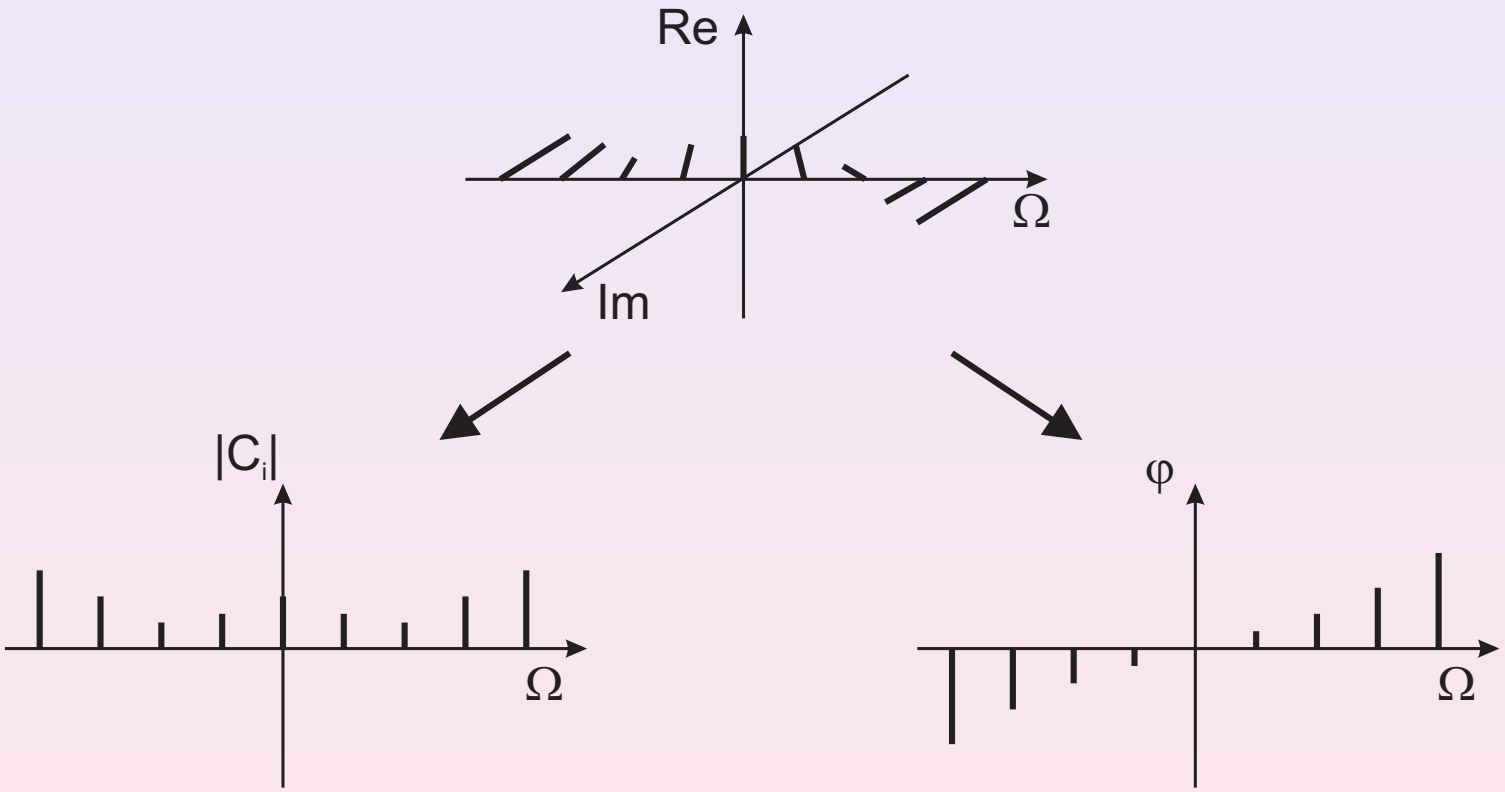
$$f(t) = \sum_{i=-\infty}^{\infty} C_i e^{ji\Omega t}$$

Koeficienti C_i so določeni z izrazom:

$$C_i(j\Omega) = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) e^{-ji\Omega t} dt$$

Imenujemo jih spektralna amplituda.

Grafični prikaz koeficientov $C_i(j\Omega)$



Velja: $\Delta\Omega = \Omega_{i+1} - \Omega_i = \dfrac{2\pi(i+1)}{T} - \dfrac{2\pi i}{T} = \dfrac{2\pi}{T} = \Omega$

Fourier-jev integral - 1

V kolikor imamo opravka z aperiodskimi funkcijami, limitiramo meje integriranja proti neskončnosti.

$$f(t) = \lim_{T\to\infty}\left[\sum_{i=-\infty}^{\infty}\frac{e^{ji\Omega t}}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)e^{-ji\Omega t}dt\right]$$

Fourier-jev integral - 2

Ker velja: $T \to \infty \Rightarrow C_i(j\Omega) \to 0$, pojem spektralne amplitude zgubi pomen. Uvedemo spektralno gostoto:

$$F' = \frac{C_i}{\Delta\Omega}$$

Fourier-jev integral - 3

Velja:

$$f(t) = \lim_{\substack{\Delta\Omega \to 0 \\ T \to \infty}} \sum_{i=-\infty}^{\infty} \frac{C_i}{\Delta\Omega} e^{ji\Omega t} \Delta\Omega = \lim_{\Delta\Omega \to 0} \sum_{\omega=-\infty}^{\infty} F'(j\omega) e^{j\omega t} \Delta\Omega$$

Z upoštevanjem $\Delta\Omega \to d\omega$ dobimo Fourier-jev integral:

$$f(t) = \int_{-\infty}^{\infty} F'(j\omega) e^{j\omega t} d\omega$$

Spektralno gostoto $F'(j\omega)$ dobimo iz izraza:

$$F'(j\omega) = \lim_{\substack{\Delta\Omega \to 0 \\ T \to \infty}} \frac{C_i}{\Delta\Omega} = \lim_{\substack{\Delta\Omega \to 0 \\ T \to \infty}} \frac{1}{T\Delta\Omega} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) e^{-ji\Omega t} dt$$

Fourier-jev integral - 4

Ker velja $T\Delta\Omega = 2\pi$, lahko zapišemo:

$$F'(j\omega) = \lim_{T\to\infty} \frac{1}{2\pi} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)e^{-j\omega t}dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

Če uporabimo zamenjavo $F(j\omega) = 2\pi F'(j\omega)$ dobimo:

$$\mathcal{F}[f(t)] = F(j\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

oziroma:

$$\mathcal{F}^{-1}[F(j\omega)] = f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega)e^{j\omega t}d\omega$$

Lastnosti Fourier-jeve transformacije - 1

- Množenje s konstanto: $\mathcal{F}\left[k\,f(t)\right] = k\,F(j\omega)$
- Vsota in razlika: $\mathcal{F}\left[f_1(t) \pm f_2(t)\right] = F_1(j\omega) \pm F_2(j\omega)$
- Prvi odvod: $\mathcal{F}\left[\frac{d}{dt}f(t)\right] = j\omega\,F(j\omega)$
- Drugi odvod: $\mathcal{F}\left[\frac{d^2}{dt^2}f(t)\right] = (j\omega)^2\,F(j\omega)$
- Višji odvodi: $\mathcal{F}\left[\frac{d^n}{dt^n}f(t)\right] = (j\omega)^n\,F(j\omega)$
- Integral: $\mathcal{F}\left[\int_0^t f(\tau)d\tau\right] = \frac{F(j\omega)}{j\omega}$
- Premik neodv. sprem.: $\mathcal{F}\left[f(t-t_0)\right] = e^{-j\omega t_0}F(j\omega)$
- Množenje z $e^{j\omega_0 t}$: $\mathcal{F}[e^{j\omega_0 t}f(t)] = F(j\omega - j\omega_0)$
- Sprememba čas. skale: $\mathcal{F}\left[f\left(\frac{t}{a}\right)\right] = |a|\,F(j\omega a)$

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Lastnosti Fourier-jeve transformacije - 2

- Križna korelacija: $\mathcal{F}\left[\int_{-\infty}^{\infty} f_1(\tau) f_2(t+\tau) d\tau\right] = F_1^*(j\omega) F_2(j\omega)$
- Avtokorelacija:
  $\mathcal{F}\left[\int_{-\infty}^{\infty} f_1(\tau) f_1(t+\tau) d\tau\right] = F_1^*(j\omega) F_1(j\omega) = |F_1(j\omega)|^2$
- Parseval. form.: $\int_{-\infty}^{\infty} f^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(j\omega)|^2 \, d\omega$
- Konvolucija : $\mathcal{F}\left[\int_{-\infty}^{\infty} f_1(\tau) f_2(t-\tau) d\tau\right] = F_1(j\omega) F_2(j\omega)$
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  $\mathcal{F}\left[\int_{-\infty}^{\infty} f_1(\tau)f_1(t+\tau)d\tau\right] = F_1^*(j\omega)F_1(j\omega) = |F_1(j\omega)|^2$
- Parseval. form.: $\int_{-\infty}^{\infty} f^2(t)dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(j\omega)|^2\, d\omega$
- Konvolucija : $\mathcal{F}\left[\int_{-\infty}^{\infty} f_1(\tau)f_2(t-\tau)d\tau\right] = F_1(j\omega)F_2(j\omega)$
- Konvolucija : $\mathcal{F}\left[f_1(t)f_2(t)\right] = \frac{1}{2\pi}\int_{-\infty}^{\infty} F_1(j\omega)F_2(j\eta - j\omega)d\omega$

Lastnosti Fourier-jeve transformacije - 2

- Križna korelacija: $\mathcal{F}\left[\int_{-\infty}^{\infty} f_1(\tau)f_2(t+\tau)d\tau\right] = F_1^*(j\omega)F_2(j\omega)$
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  $\mathcal{F}\left[\int_{-\infty}^{\infty} f_1(\tau)f_1(t+\tau)d\tau\right] = F_1^*(j\omega)F_1(j\omega) = |F_1(j\omega)|^2$
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Lastnosti Fourier-jeve transformacije - 3

- Simetrija: $\mathcal{F}\left[F(jt)\right] = 2\pi f(-\omega)$

Križna korelacija

Velja:

$$\mathcal{F}\left[\int_{-\infty}^{\infty} f_1(\tau) f_2(t+\tau) d\tau\right] =$$
$$\int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} f_1(\tau) f_2(t+\tau) d\tau\right] e^{-j\omega t} dt$$

Z uvedbo nove spremenljivke $u = t + \tau$ in $du = d\tau$ dobimo:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_1(\tau) f_2(u) e^{j\omega\tau} e^{-j\omega u} d\tau\, du =$$
$$\int_{-\infty}^{\infty} f_1(\tau) e^{j\omega\tau} d\tau \int_{-\infty}^{\infty} f_2(u) e^{-j\omega u} du = F_1(-j\omega)\, F_2(j\omega)$$

Avtokorelacija

Lastnost avtokorelacije dobimo, če v lastnosti križne korelacije predpostavimo $f_2(\tau) = f_1(\tau)$.

Konvolucija - 1

Velja:

$$\mathcal{F}\left[\int_{-\infty}^{\infty} f_1(\tau) f_2(t-\tau) d\tau\right] =$$
$$\int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} f_1(\tau) f_2(t-\tau) d\tau\right] e^{-j\omega t} dt$$

Z uvedbo nove spremenljivke $u = t - \tau$ in $du = dt$ dobimo:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_1(\tau) f_2(u) e^{-j\omega\tau} e^{-j\omega u} d\tau\, du =$$
$$\int_{-\infty}^{\infty} f_1(\tau) e^{-j\omega\tau} d\tau \int_{-\infty}^{\infty} f_2(u) e^{-j\omega u} du = F_1(j\omega)\, F_2(j\omega)$$

Konvolucija - 2

Velja:

$$\mathcal{F}\left[f_1(t)f_2(t)\right] = \int_{-\infty}^{\infty} f_1(t)f_2(t)e^{-j\eta t}dt$$

Če za $f_1(t)$ vstavimo $f_1(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F_1(j\omega)e^{j\omega t}d\omega$, dobimo:

$$\mathcal{F}\left[f_1(t)f_2(t)\right] = \int_{-\infty}^{\infty} \frac{1}{2\pi}\int_{-\infty}^{\infty} F_1(j\omega)e^{j\omega t}d\omega\, f_2(t)e^{-j\eta t}dt =$$

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} F_1(j\omega)\int_{-\infty}^{\infty} f_2(t)e^{-j(\eta-\omega)t}dt\, d\omega =$$

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} F_1(j\omega)F_2(j\eta - j\omega)d\omega$$

Parsevalova identiteta

Konvolucija za $f_1(t) = f_2(t) = f(t)$:

$$\mathcal{F}\left[f(t)f(t)\right] = \int_{-\infty}^{\infty} f^2(t)e^{-j\eta t}dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\omega)F(j\eta - j\omega)d\omega$$

Če v zgornji izraz vstavimo $\eta = 0$ in upoštevamo enakost $F(-j\omega)F(j\omega) = |F(j\omega)|^2$, dobimo:

$$\int_{-\infty}^{\infty} f^2(t)dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(j\omega)|^2 d\omega$$

Simetrija

Predpostavimo, da velja $g(t) = F(jt)$:

$$G(j\omega) = \int_{-\infty}^{\infty} g(t)e^{-j\omega t}dt = \int_{-\infty}^{\infty} F(jt)\, e^{-j\omega t}dt = 2\pi\left(\frac{1}{2\pi}\int_{-\infty}^{\infty} F(jt)\, e^{j(-\omega)t}dt\right)$$

Ker je faktor v oklepaju ravno inverzna Fourier-jeva transformacija funkcije F(jt), pri parametru $-\omega$, dobimo:

$$\mathcal{F}\left[F(jt)\right] = G(j\omega) = 2\pi\, f(-\omega)$$

Fourier-jeve slike nekaterih pomembnejših funkcij

$f(t)$	$F(j\omega)$
$\delta(t)$	1
1	$\pi\,\delta(\omega) + \dfrac{1}{j\omega}$
$e^{-at}\,u(t),\ a > 0$	$\dfrac{1}{j\omega + a}$
$\sin(\omega_0 t)$	$\dfrac{\pi}{j}\left[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)\right]$
$\cos(\omega_0 t)$	$\pi\ \left[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)\right]$
$\sum_{i=-\infty}^{\infty} \delta(t - iT)bt$	$\dfrac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \dfrac{2\pi k}{T}\omega_0\right)$

Fourier-jeva slika enotske impulzne (delta) funkcije

Izhodišče:

$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt$

Ker je delta funkcija različna od nič le, ko velja $t = 0$, kjer je $e^{-j\omega t} = 1$ in ker hkrati velja $\int_{-\infty}^{\infty} \delta(t) dt = 1$, lahko zapišemo:

$\mathcal{F}[\delta(t)] = 1$

Fourier-jeva slika enotske skočne funkcije - 1

Najprej določiti Fourierjeve sliko funkcije $\mathrm{sign}(t)$, ki je deinirana z:

$$\mathrm{sign}(t) = \begin{cases} -1; & t < 0 \\ 1; & t \geq 0 \end{cases}$$

Njena povezava z enotsko koračno funkcijo $u(t)$ je dana z izrazom:

$$\mathrm{sign}(t) = 2\,u(t) - 1$$

Če ta izraz odvajamo, dobimo:

$$\frac{d}{dt}\mathrm{sign}(t) = 2\delta(t)$$

Fourier-jeva slika enotske skočne funkcije - 2

Če Fourier-jevo transformacijo funkcije $\mathrm{sign}(t)$ označimo z $X(j\omega)$, mora veljati:

$$j\omega X(j\omega) = \mathcal{F}\left[2\delta(t)\right] = 2 \quad \Rightarrow \quad X(j\omega) = \frac{2}{j\omega}$$

Drugi korak pri določitvi Fourier-jeve transformacije enotske skočne funkcije je določitev Fourier-jeve transformacije funkcije $f(t) = 1$. Ker je Fourier-jeva transformacije enotske impulzne funkcije enaka $\mathcal{F}\delta(t) = 1$, velja (lastnost simetrije):

$$\mathcal{F}\left[1\right] = 2\pi\delta(-\omega) = 2\pi\delta(\omega)$$

Fourier-jeva slika enotske skočne funkcije - 3

V nadaljevanju bomo rezultata dobljena v prvih dveh korakih uporabili. Enotsko skočno funkcijo namreč lahko z uporabo funkcije $\mathrm{sign}(t)$ zapišemo v obliki:

$$u(t) = \frac{1}{2} + \frac{1}{2}\mathrm{sign}(t) = \frac{1}{2}\left(1 + \mathrm{sign}(t)\right)$$

Z upoštevanjem linearnosti Fourier-jeve transformacije dobimo:

$$\mathcal{F}\left[u(t)\right] = \frac{1}{2}\left(\mathcal{F}\left[1\right] + \mathcal{F}\left[\mathrm{sign}(t)\right]\right) = \pi\delta(\omega) + \frac{1}{j\omega}$$

Frekvenčna karakteristika 1

Izhodišče:

$$\frac{a_n d^n y(t)}{dt^n} + \cdots + a_1 \frac{dy(t)}{dt} + a_0 y(t) =$$
$$b_m \frac{d^m x(t)}{dt^m} + \cdots + b_1 \frac{dx(t)}{dt} + b_0 x(t)$$

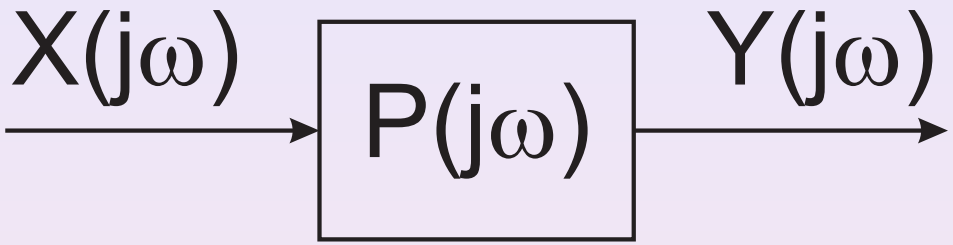
Upoštevamo lastnosti Fourier-jeve transformacije:

$$\left(a_n (j\omega)^n + \cdots + a_1 j\omega + a_0\right) Y(j\omega) =$$
$$\left(b_m (j\omega)^m + \cdots + b_1 j\omega + b_0\right) X(j\omega)$$

Z ureditvijo dobimo frekv. karakt. $P(j\omega)$:

$$P(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{b_m (j\omega)^m + \cdots + b_1 j\omega + b_0}{a_n (j\omega)^n + \cdots + a_1 j\omega + a_0}$$

Frekvenčna karakteristika 2



Blok je grafični zapis za izraz:

$$Y(j\omega) = P(j\omega) \cdot X(j\omega)$$

Druga možnost zapisa $P(j\omega)$:

$$P(j\omega) = |P(j\omega)|e^{j\angle P(j\omega)} = |P(j\omega)|e^{j\varphi}$$

kjer velja:

$$|P(j\omega)| = \sqrt{(\mathrm{Re}P(j\omega))^2 + (\mathrm{Im}P(j\omega))^2}$$

$$\angle P(j\omega) = \varphi = \arctan \frac{\mathrm{Im}P(j\omega)}{\mathrm{Re}P(j\omega)}$$

Frekvenčna karakteristika - Sinusni signal - 1

Vstopni signal je sinusni signal z amplitudo 1:

$X(j\omega) = \mathcal{F}\left[\sin(\omega_0 t)\right] = \frac{\pi}{j}\left[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)\right]$

Fourierjeva slika izstopnega signala:

$$Y(j\omega) = P(j\omega)\,X(j\omega) = P(j\omega)\frac{\pi}{j}\left[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)\right]$$

Signal $y(t)$ dobimo z inverzno Fourier-jevo transformacijo:

$$y(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{j\omega t} Y(j\omega) d\omega =$$

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{j\omega t} P(j\omega)\frac{\pi}{j}\left[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)\right] d\omega$$

Frekvenčna karakteristika - Sinusni signal - 2

Če za $P(j\omega)$ vstavimo $|P(j\omega)|e^{j\angle P(j\omega)}$ in upoštevamo lastnosti $\delta(t)$ funkcije, dobimo:

$$y(t) = \frac{1}{2j}\left[|P(j\omega_0)|e^{j\angle P(j\omega_0)}e^{j\omega_0 t} - |P(-j\omega_0)|e^{j\angle P(-j\omega_0)}e^{-j\omega_0 t}\right]$$

Ker velja $|P(-j\omega_0)| = |P(j\omega_0)|$ in $\angle P(-j\omega) = -\angle P(j\omega)$, lahko zapišemo:

$$y(t) = \frac{1}{2j}|P(j\omega_0)|\left[e^{j(\omega_0 t + \angle P(j\omega_0))} - e^{-j(\omega_0 t + \angle P(j\omega_0))}\right]$$

Z upoštevanjem Eulerjeve formule $sin(\alpha) = \frac{e^{j\alpha} - e^{-j\alpha}}{2j}$, dobimo:

$$y(t) = |P(j\omega_0)|\sin\left(\omega_0 t + \angle P(j\omega_0)\right) = |P(j\omega_0)|\sin\left(\omega_0 t + \varphi\right)$$

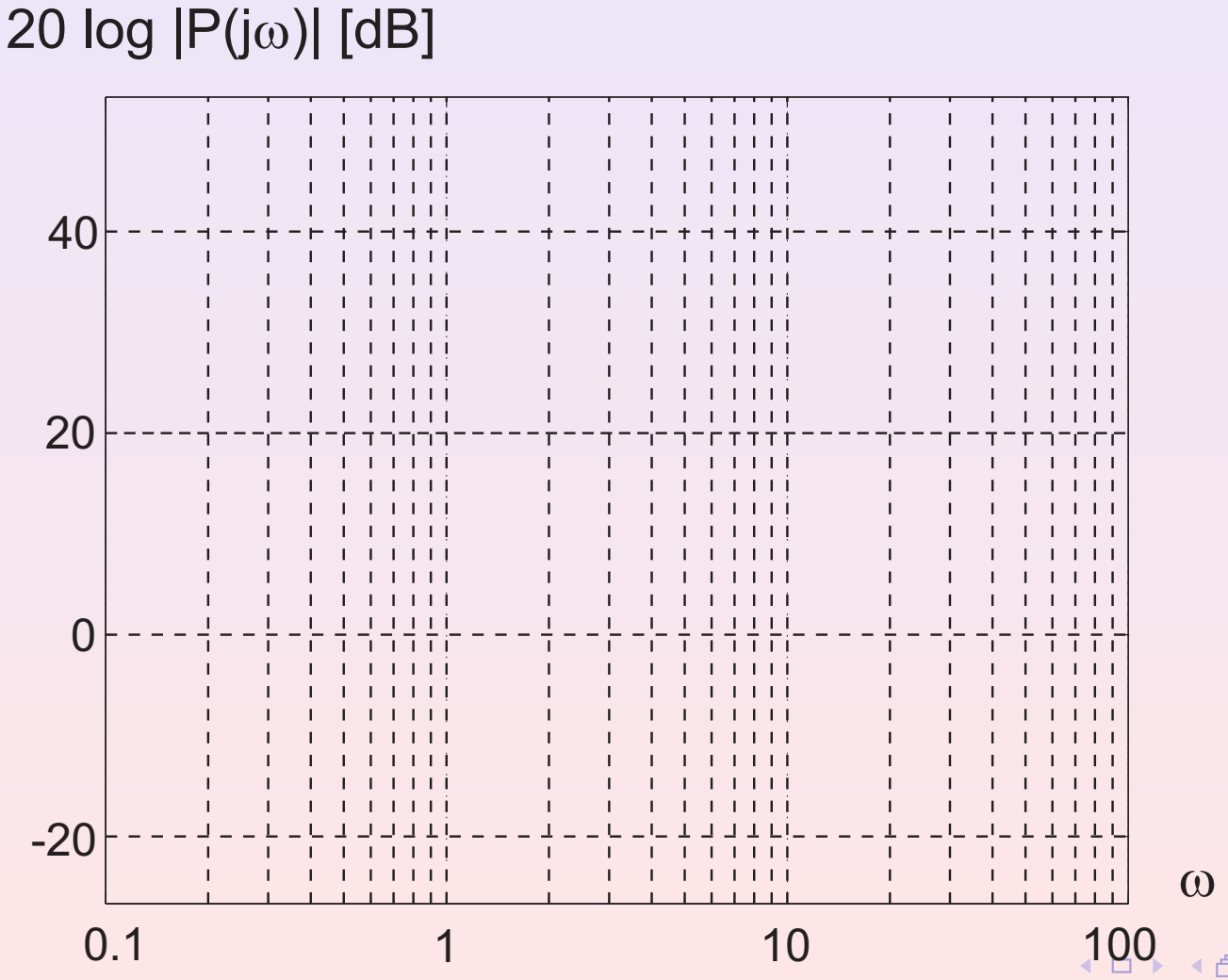
Frekvenčna karakteristika - Zaključek

$$X \sin(\omega t) \rightarrow X e^{j\omega t} \rightarrow P(j\omega) \rightarrow Y e^{j(\omega t+\varphi)} \rightarrow Y \sin(\omega t+\varphi)$$

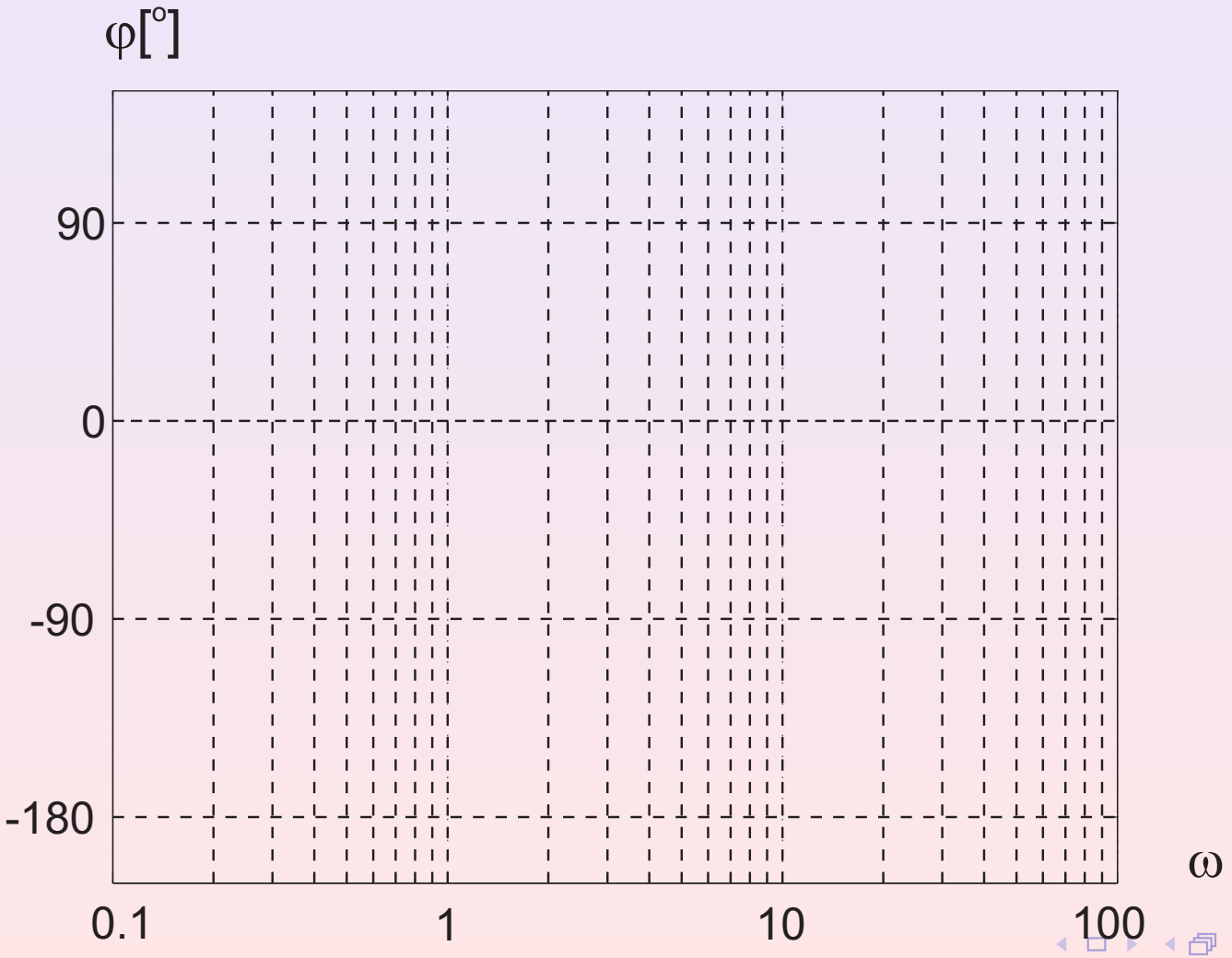
Frekvenčni diagrami - Uvod

Frekvenčni diagrami so grafi, ki prikazujejo odvisnost med razmerjem amplitud izstopnega in vstopnega signala in fazo kot odvisnima spremenljivkama ter frekvenco kot neodvisno spremenljivko. Najbolj znana sta Bode-jev in Nyquist-ov diagram, ki bosta podrobneje predstavljena. Manj znan je prirodni diagram. Čeprav ga je enostavno narisati, je njegova uporabna vrednost manjša od predhodno omenjenih frekvenčnih diagramov.

Bodejev diagram

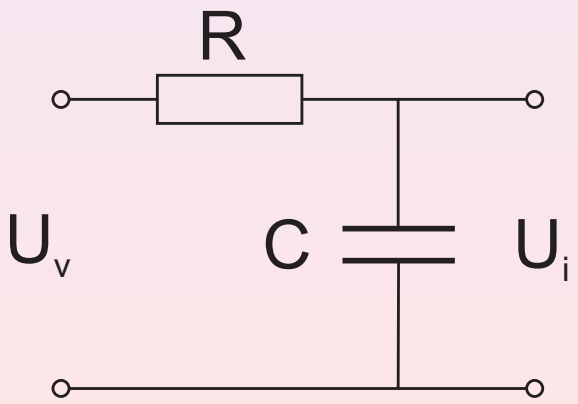


Bodejev diagram



Bodejev diagram - Primer - 1

Nariši Bodejev diagram za sistem prikazan na sliki. Sistem sestavljata upornik z upornostjo $R = 100k\Omega$ in kondenzator s kapacitivnostjo $C = 10\mu F$. Na osnovi Bodejevega diagrama določi odziv sistema, če nanj delujemo z vstopno napetostjo $U_v(t) = 5\sin(0.1t)V$ in če nanj delujemo z napetostjo $U_v(t) = 5\sin(10t)V$.



Bodejev diagram - Primer - 2

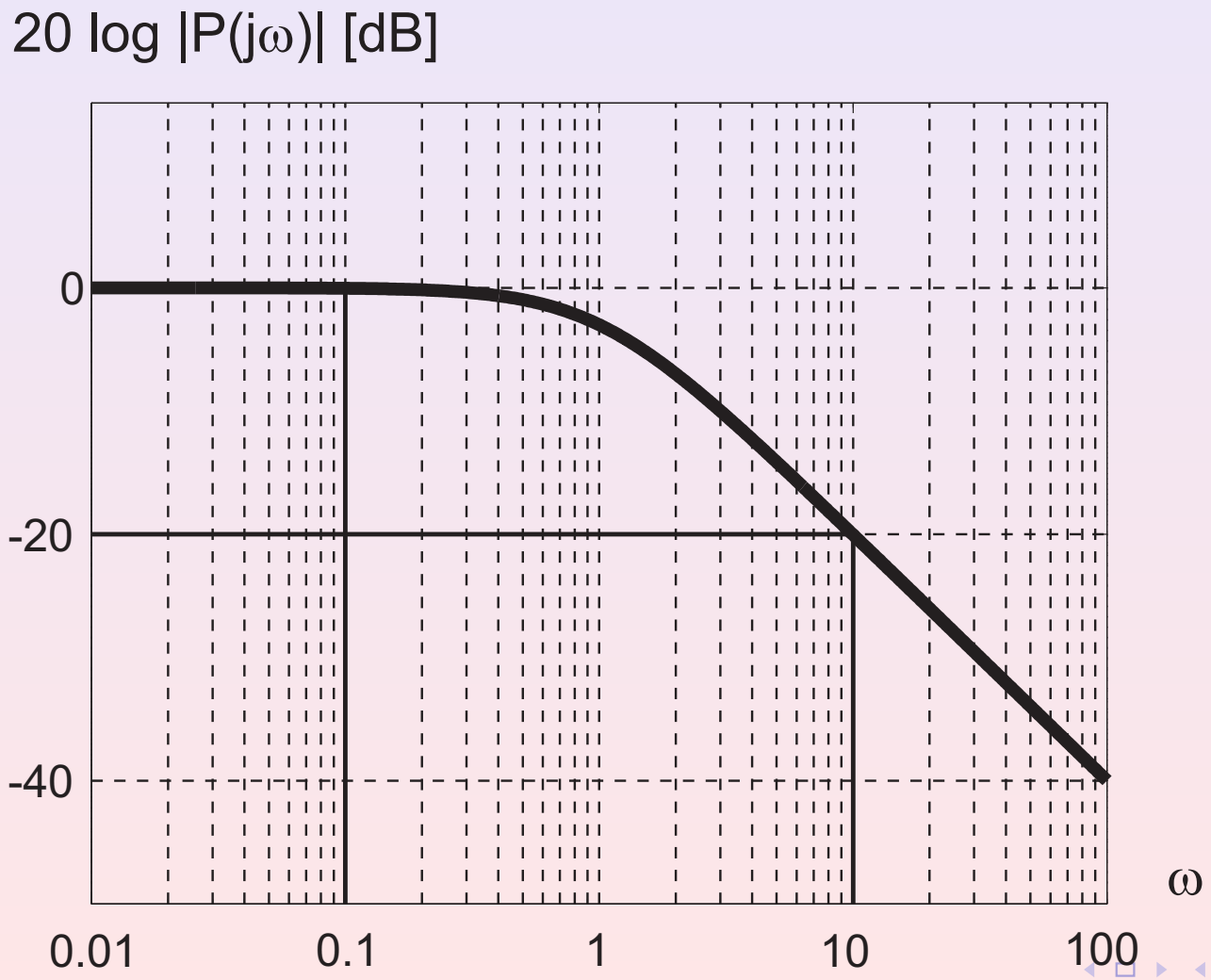
Prenosna karakteristika:

$$\frac{U_i(t)}{U_v(t)} = \frac{\frac{1}{CD}}{R + \frac{1}{CD}} = \frac{1}{RCD + 1}$$

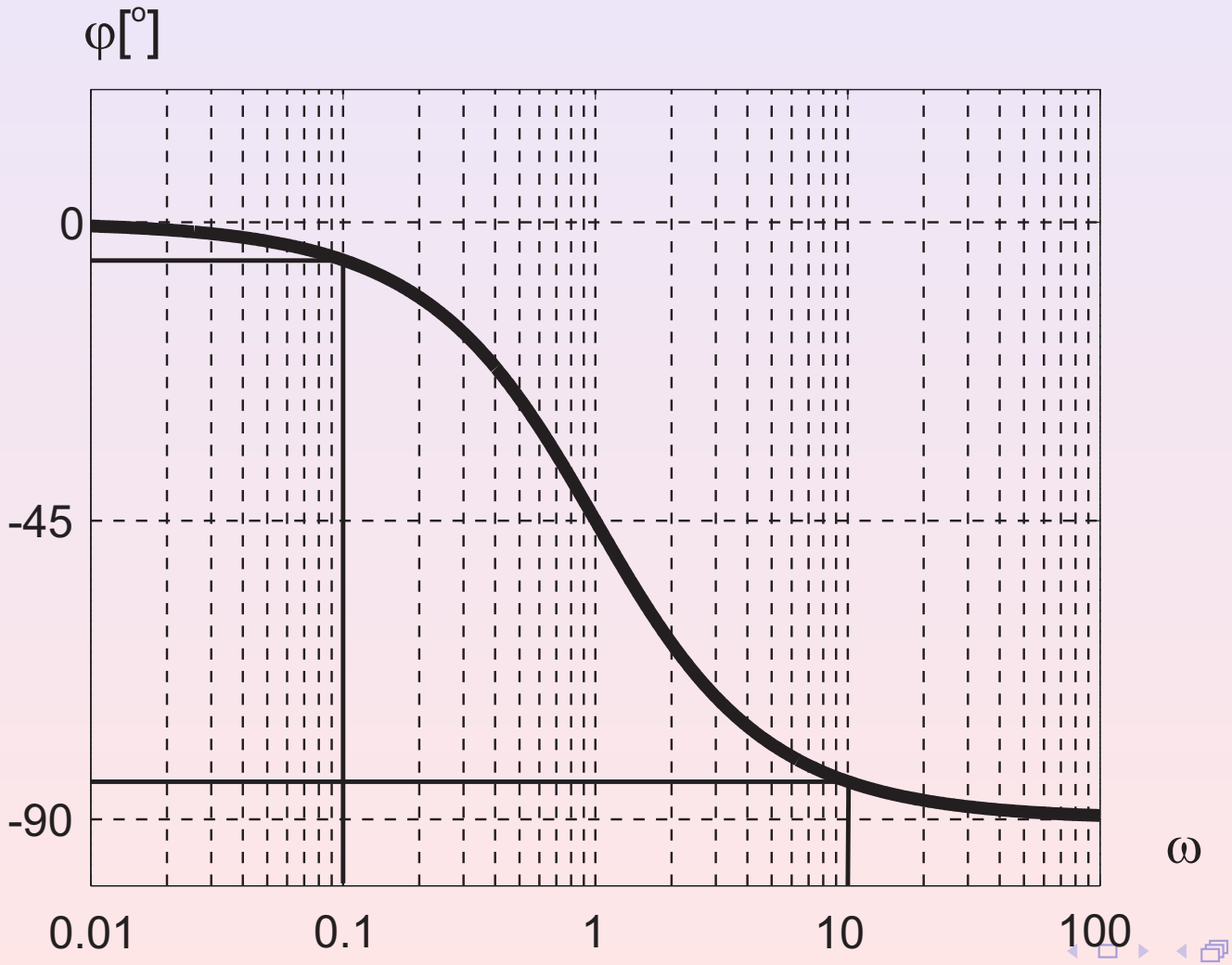
Frekvenčna karakteristika:

$$P(j\omega) = \frac{1}{RC\,j\omega + 1} = \frac{1}{j\omega + 1}$$

Bodejev diagram - Primer - 3 (Grafa)



Bodejev diagram - Primer - 3 (Grafa)



Bodejev diagram - Primer - 4 ($\omega_1 = 0.1\mathrm{rad/s}$)

Iz odčitka dobimo:

$$20\log\frac{|Y(j\omega_1)|}{|X(j\omega_1)|} = 20\log\frac{|Y(j\omega_1)|}{5} = -0.0432 \Rightarrow |Y(j\omega_1)| = 5\cdot 10^{\frac{-0.0432}{20}} = 4.9752 \approx 5$$

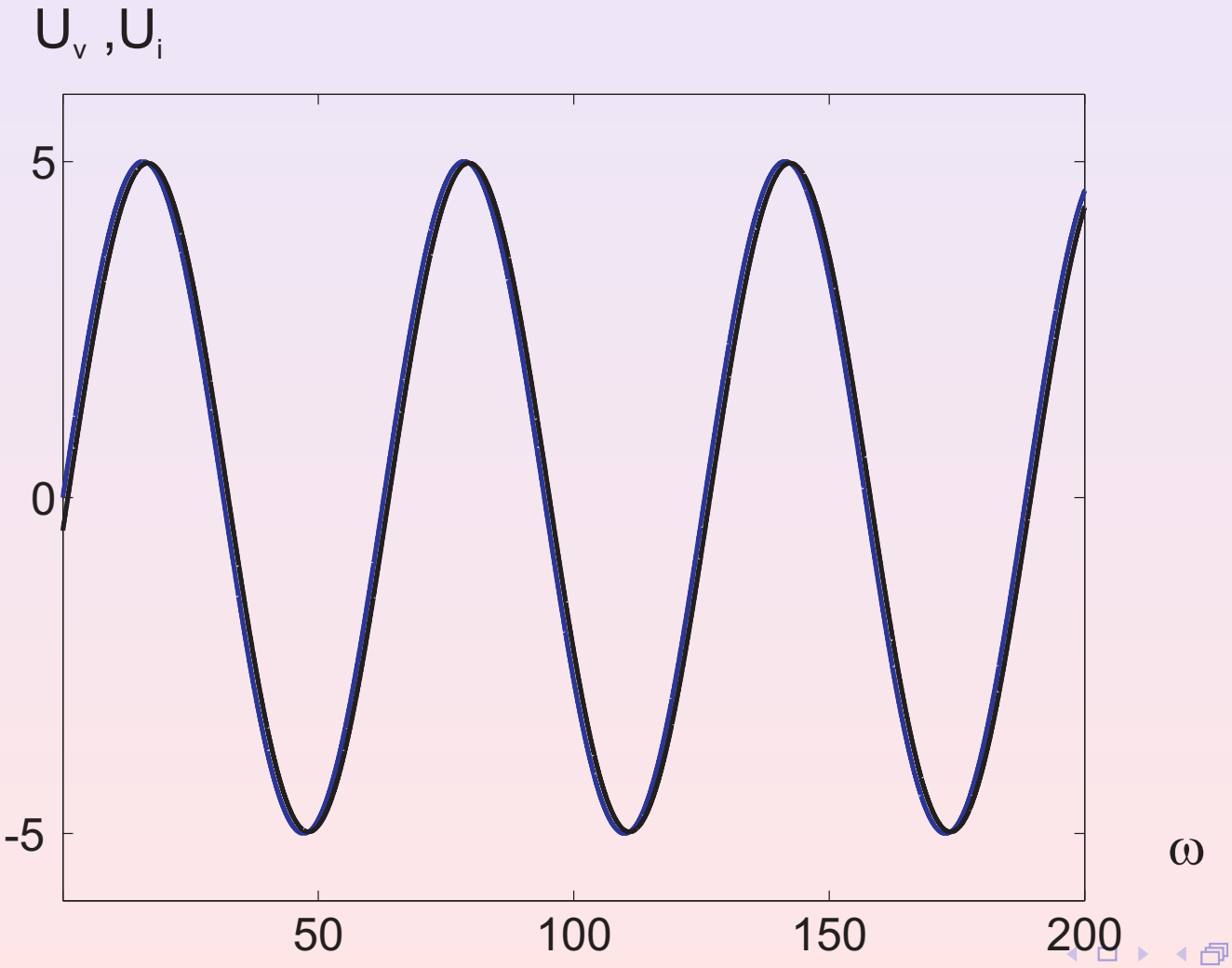
Faza:

$$\varphi_1 = -5.7106\,^\circ$$

Izstopni signal:

$$U_i(t) = 4.9752\sin(0.1t - 5.7106\,^\circ)V$$

Bodejev diagram - Primer - 5 (Grafa)



Bodejev diagram - Primer - 6 ($\omega_2 = 10\mathrm{rad/s}$)

Iz odčitka dobimo:

$$20\log\frac{|Y(j\omega_2)|}{|X(j\omega_2)|} = 20\log\frac{|Y(j\omega_2)|}{5} = -20.0432 \Rightarrow |Y(j\omega_2)| = 5\cdot 10^{\frac{-20.0432}{20}} = 0.4975$$

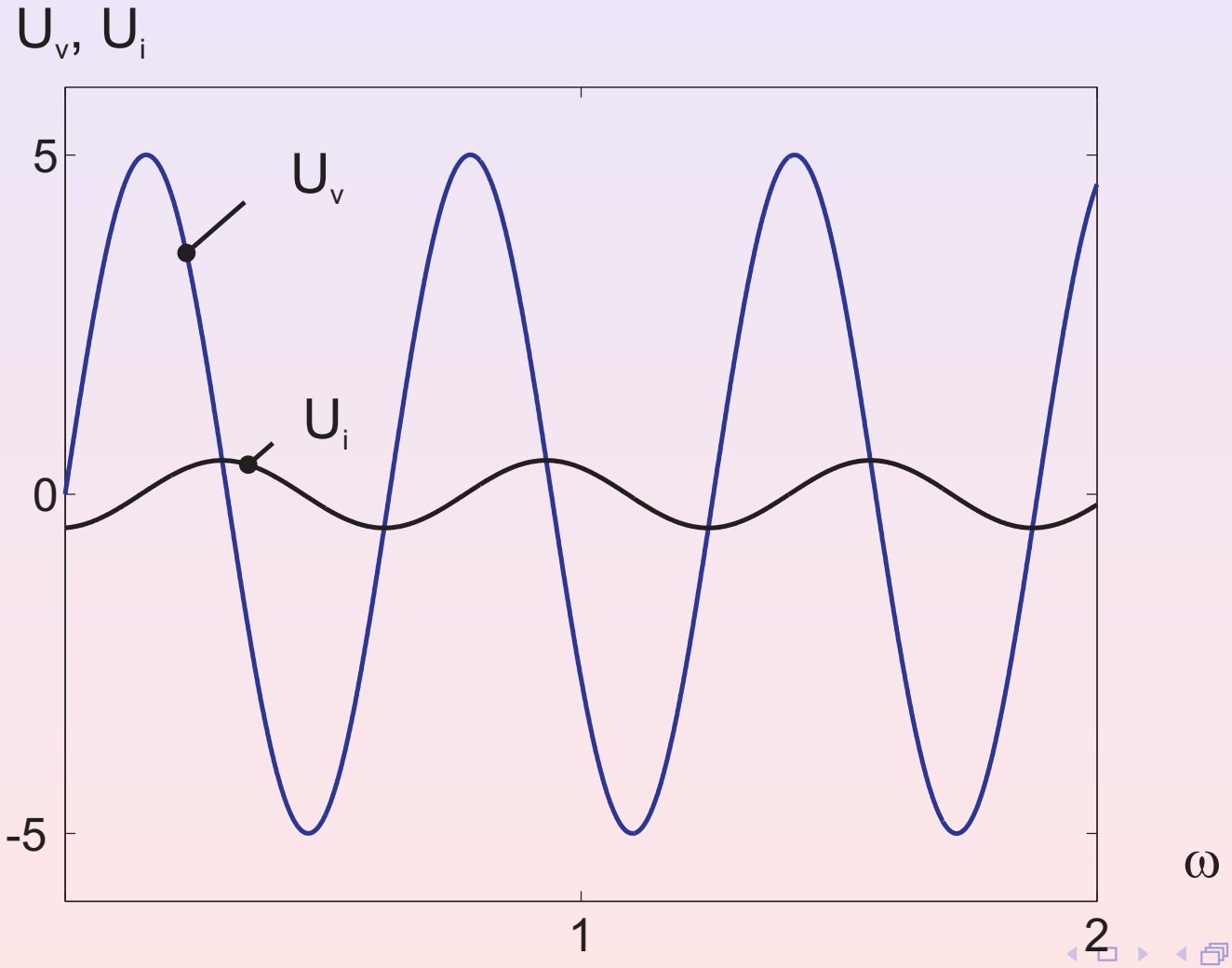
Faza:

$$\varphi_2 = -84.2894\,^\circ$$

Izstopni signal:

$$U_i(t) = 0.4975\sin(10t - 84.2894\,^\circ)V$$

Bodejev diagram - Primer - 7 (Grafa)

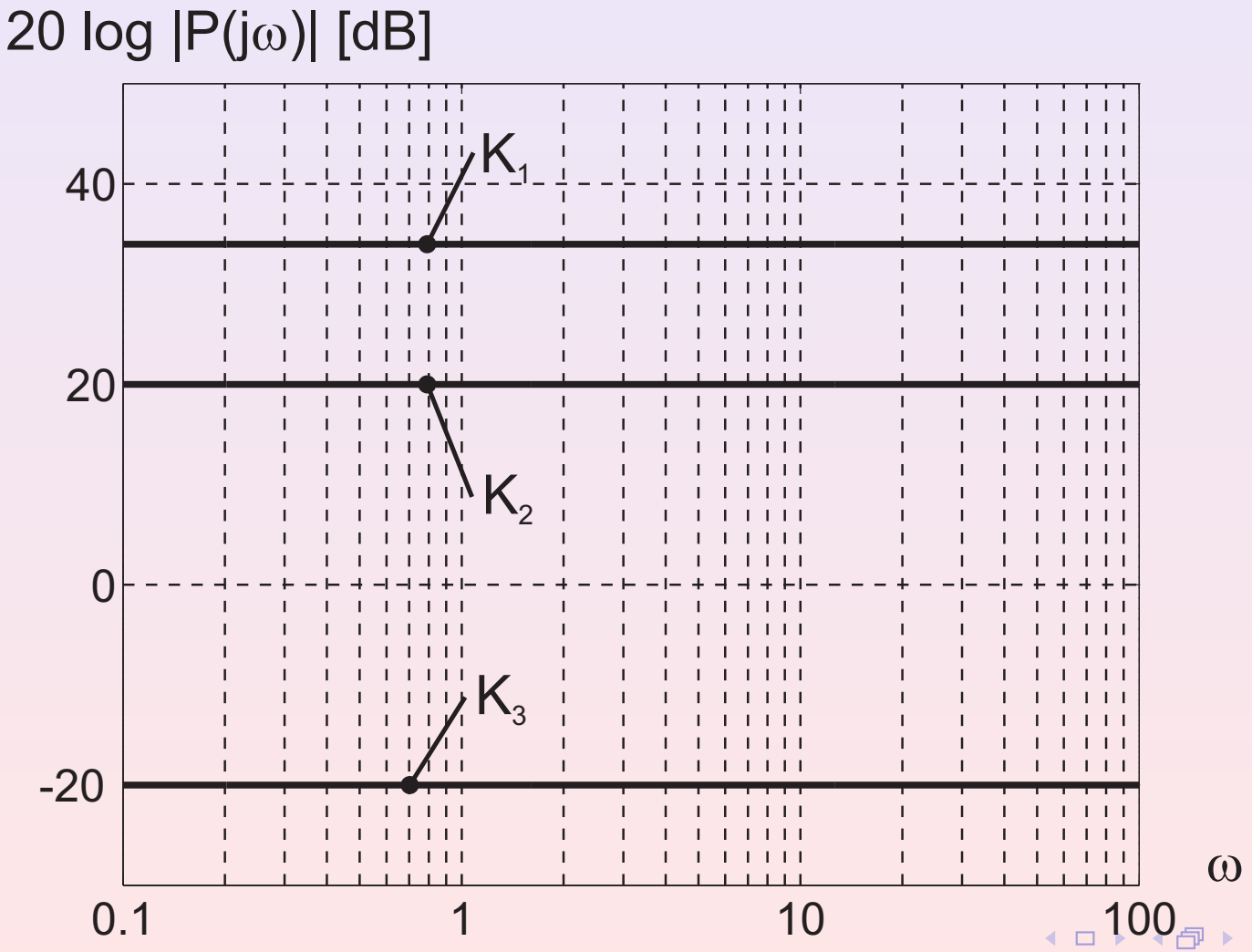


Splošni Bodejev diagram

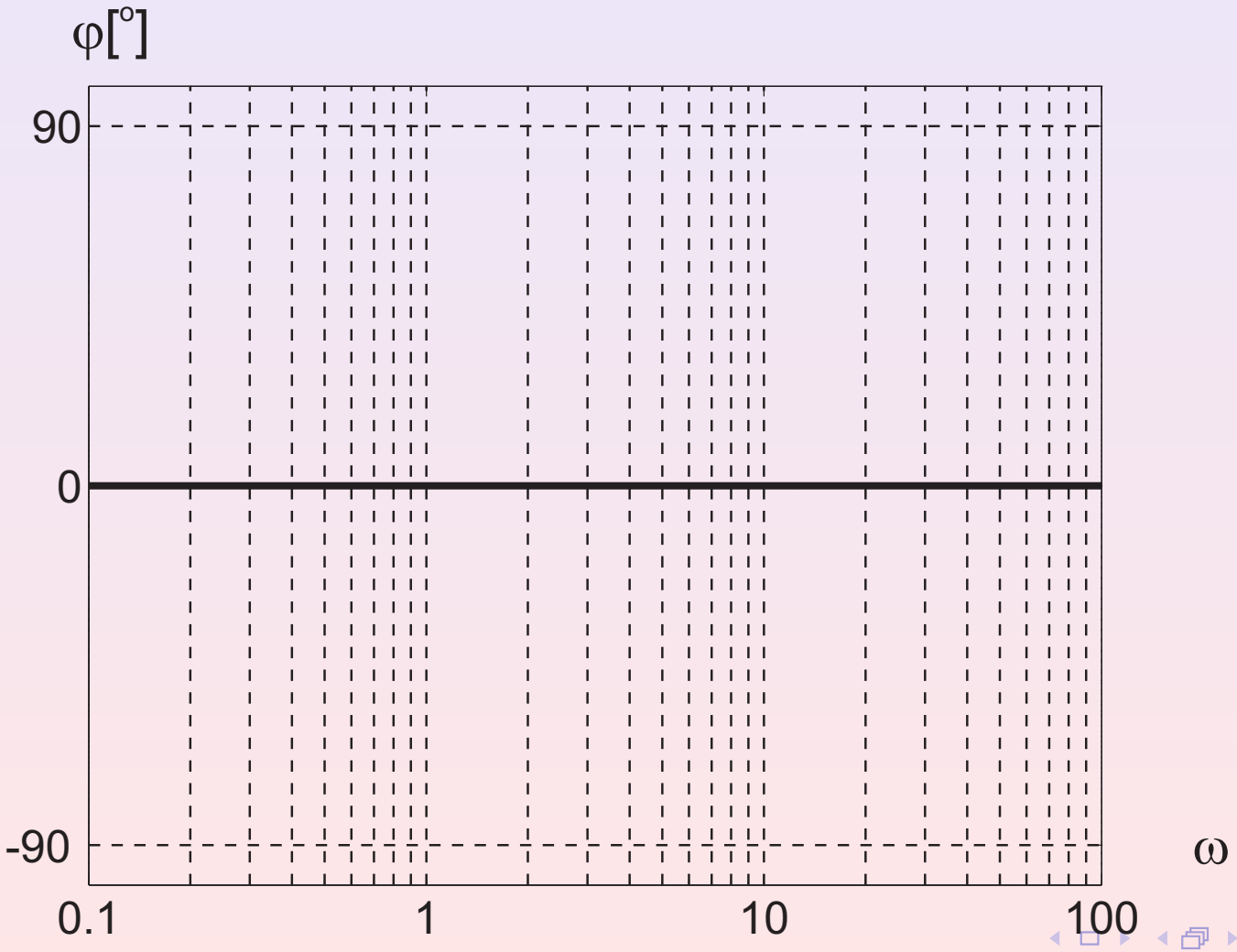
Izhodišče:

$$P(j\omega) = \frac{K \prod_{i=1}^{Q}(1 + j\omega T_i) \prod_{k=1}^{M}\left(1 + \frac{2\zeta_k}{\omega_{nk}} j\omega + \left(\frac{j\omega}{\omega_{nk}}\right)^2\right)}{(j\omega)^N \prod_{h=1}^{P}(1 + j\omega T_h) \prod_{l=1}^{R}\left(1 + \frac{2\zeta_l}{\omega_{nl}} j\omega + \left(\frac{j\omega}{\omega_{nl}}\right)^2\right)}$$

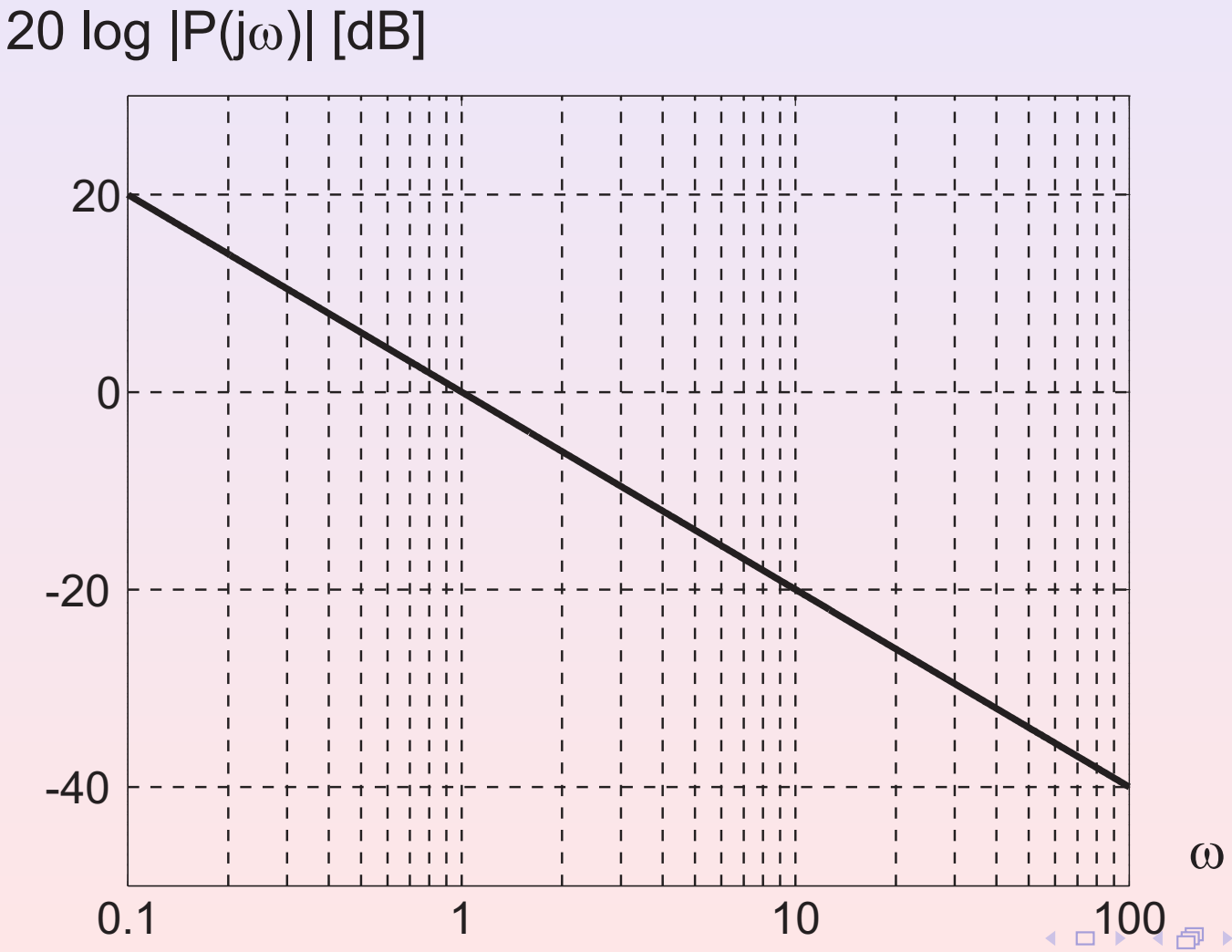
Grafi



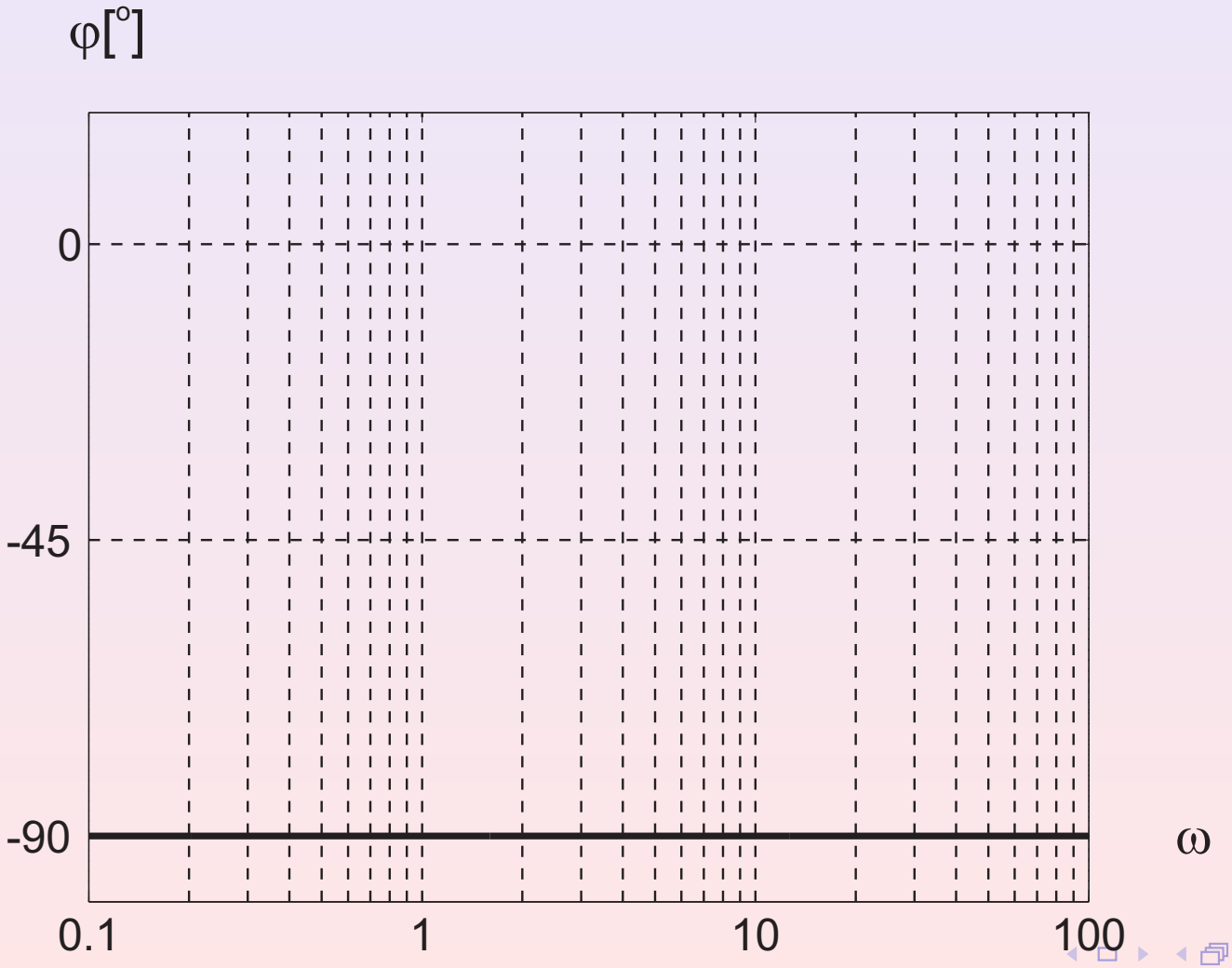
Grafi



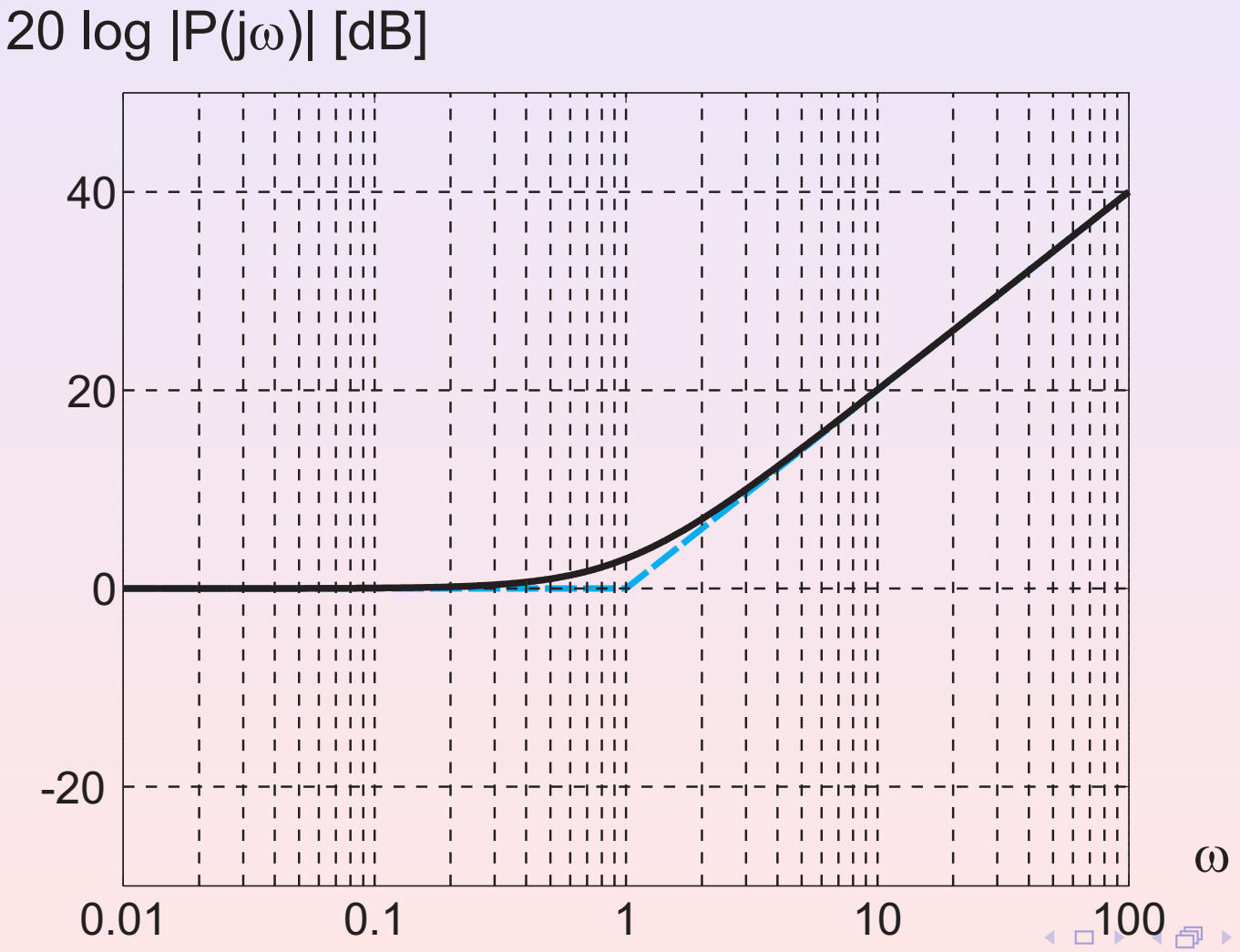
Grafi



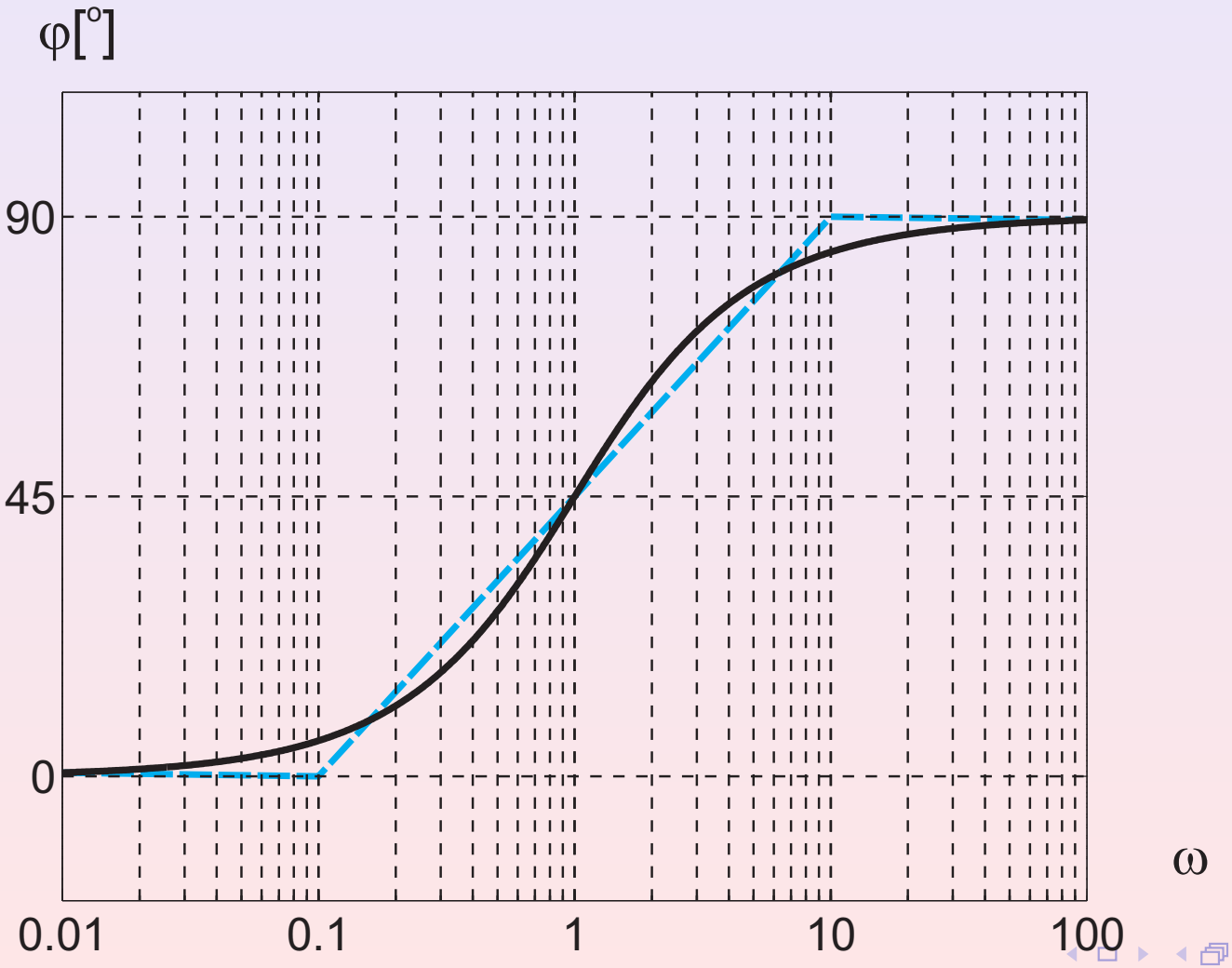
Grafi



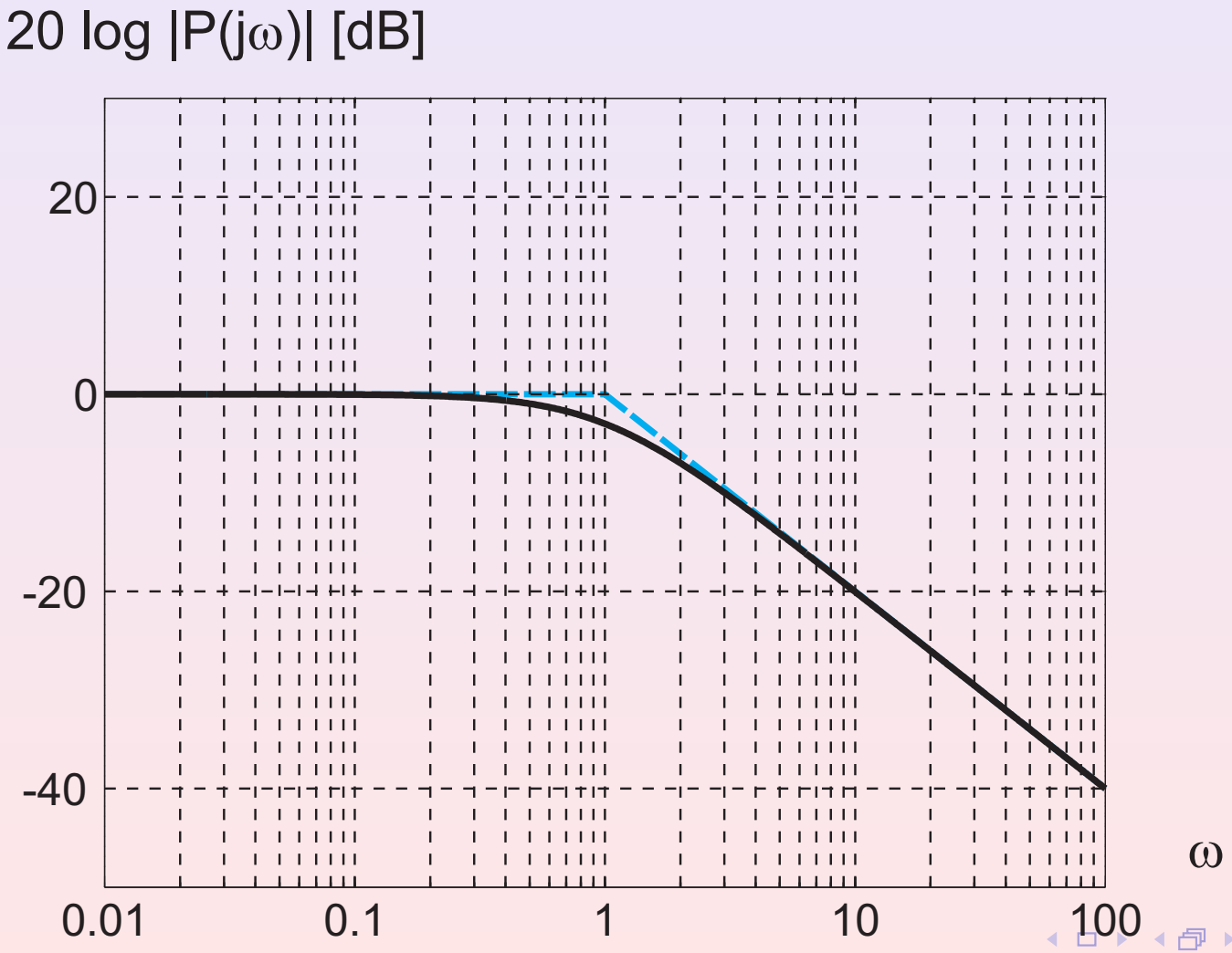
Grafi



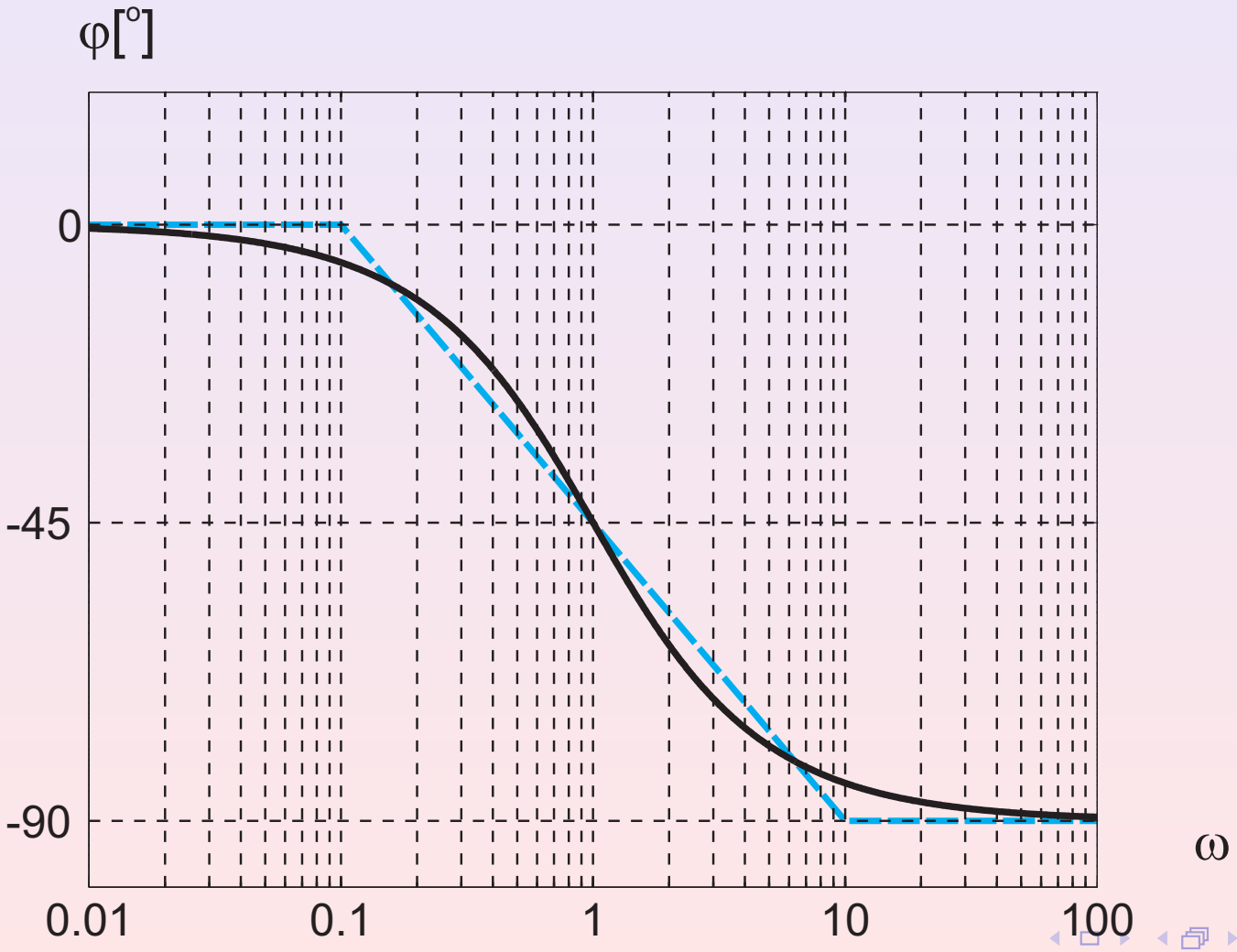
Grafi



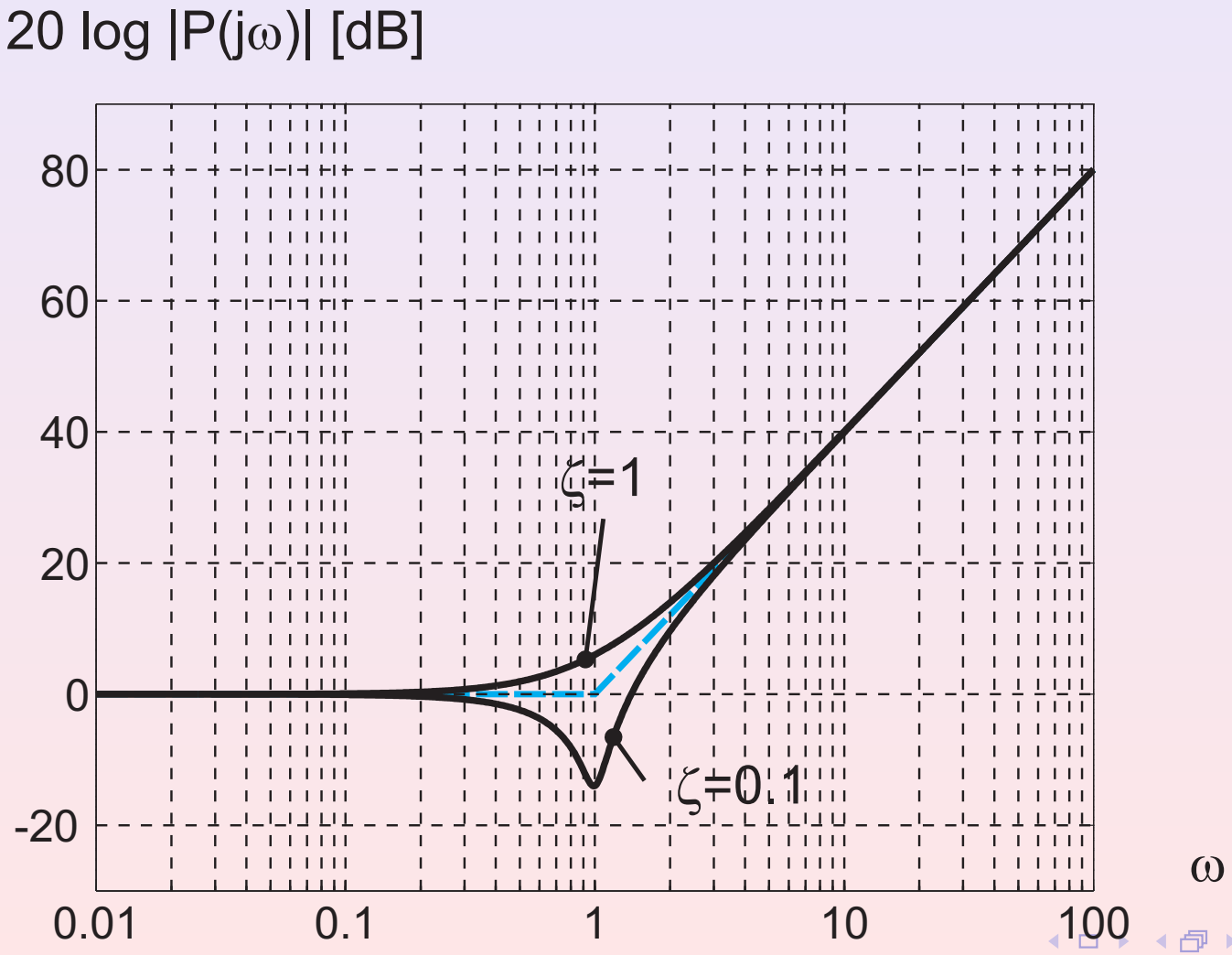
Grafi



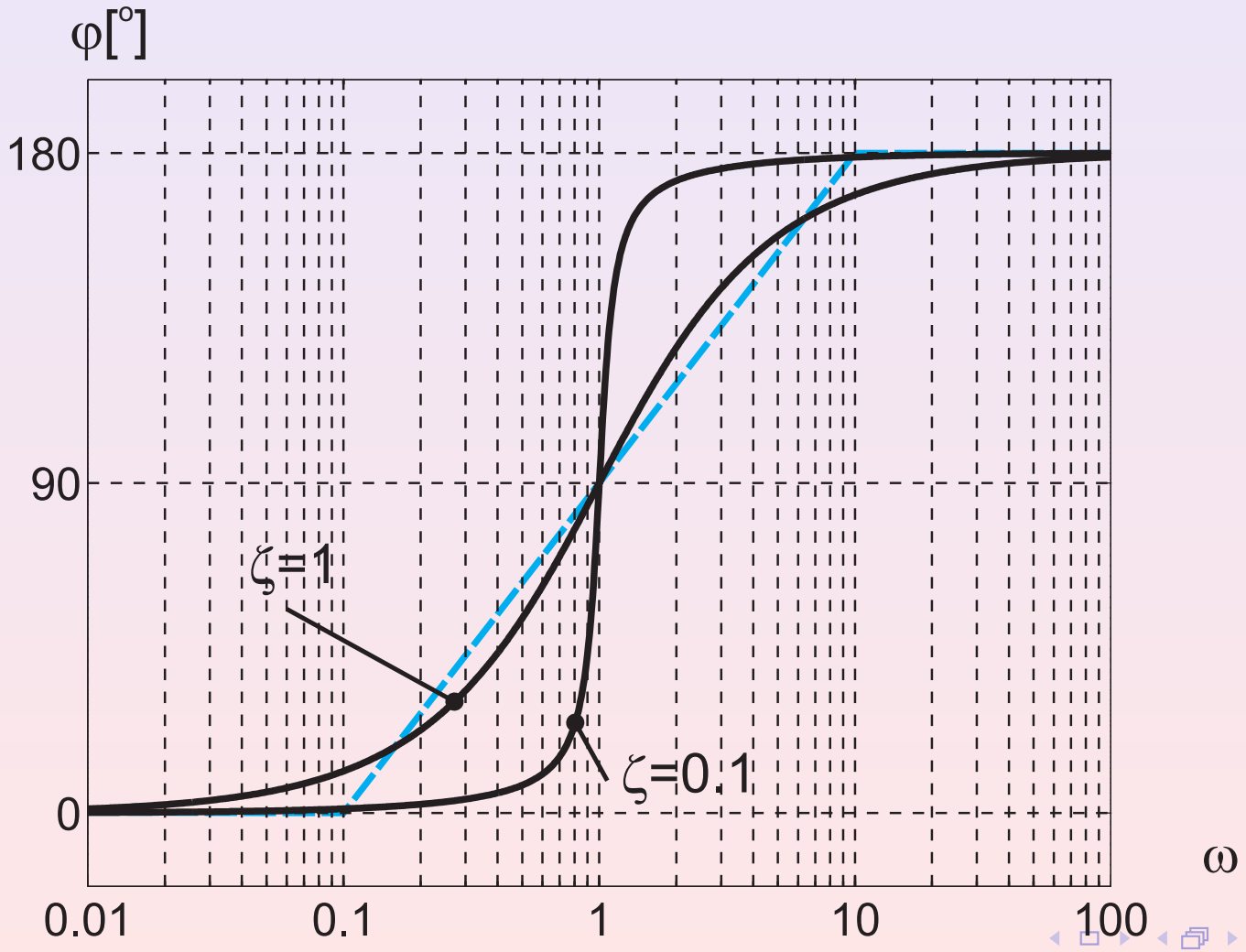
Grafi



Grafi

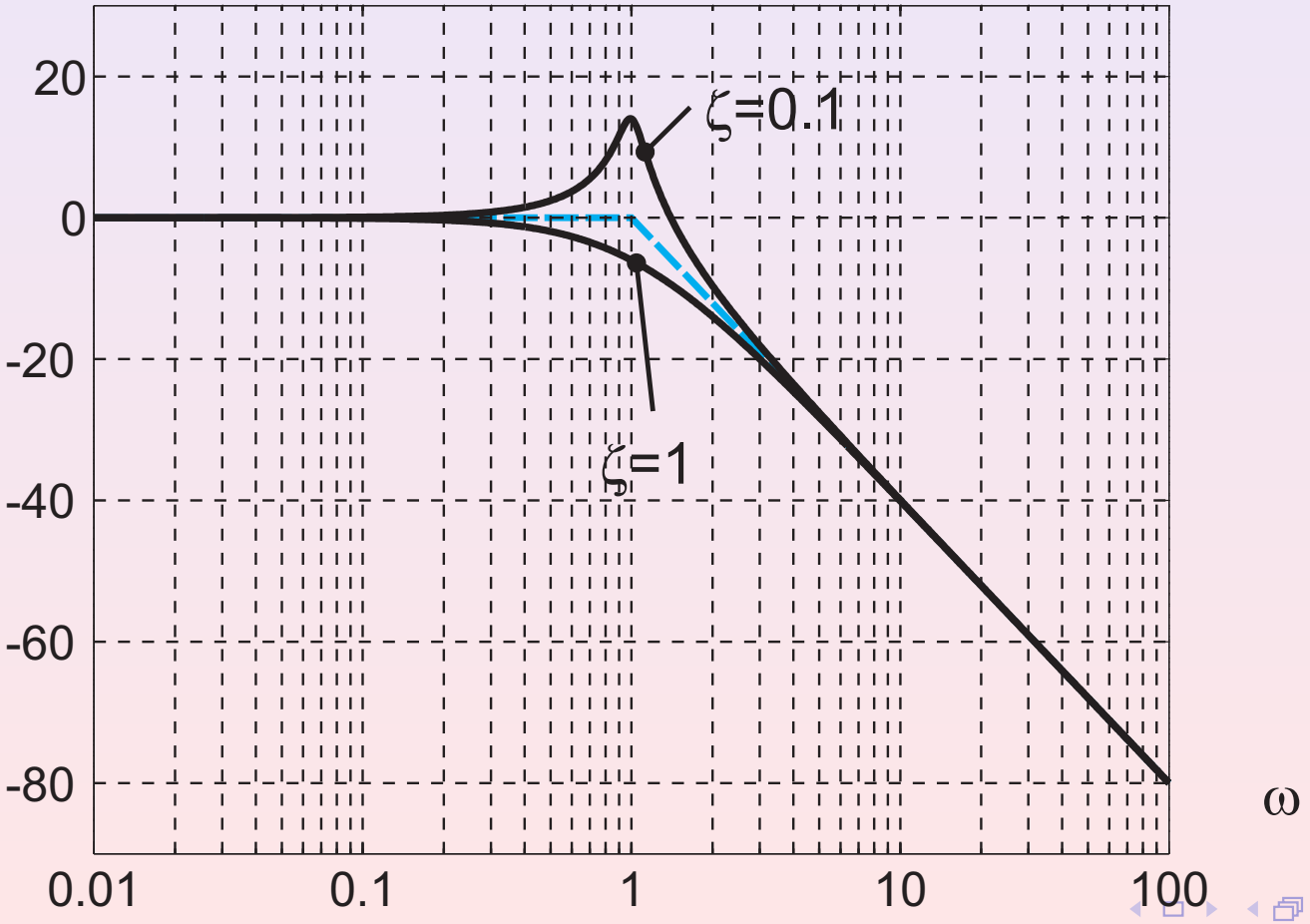


Grafi

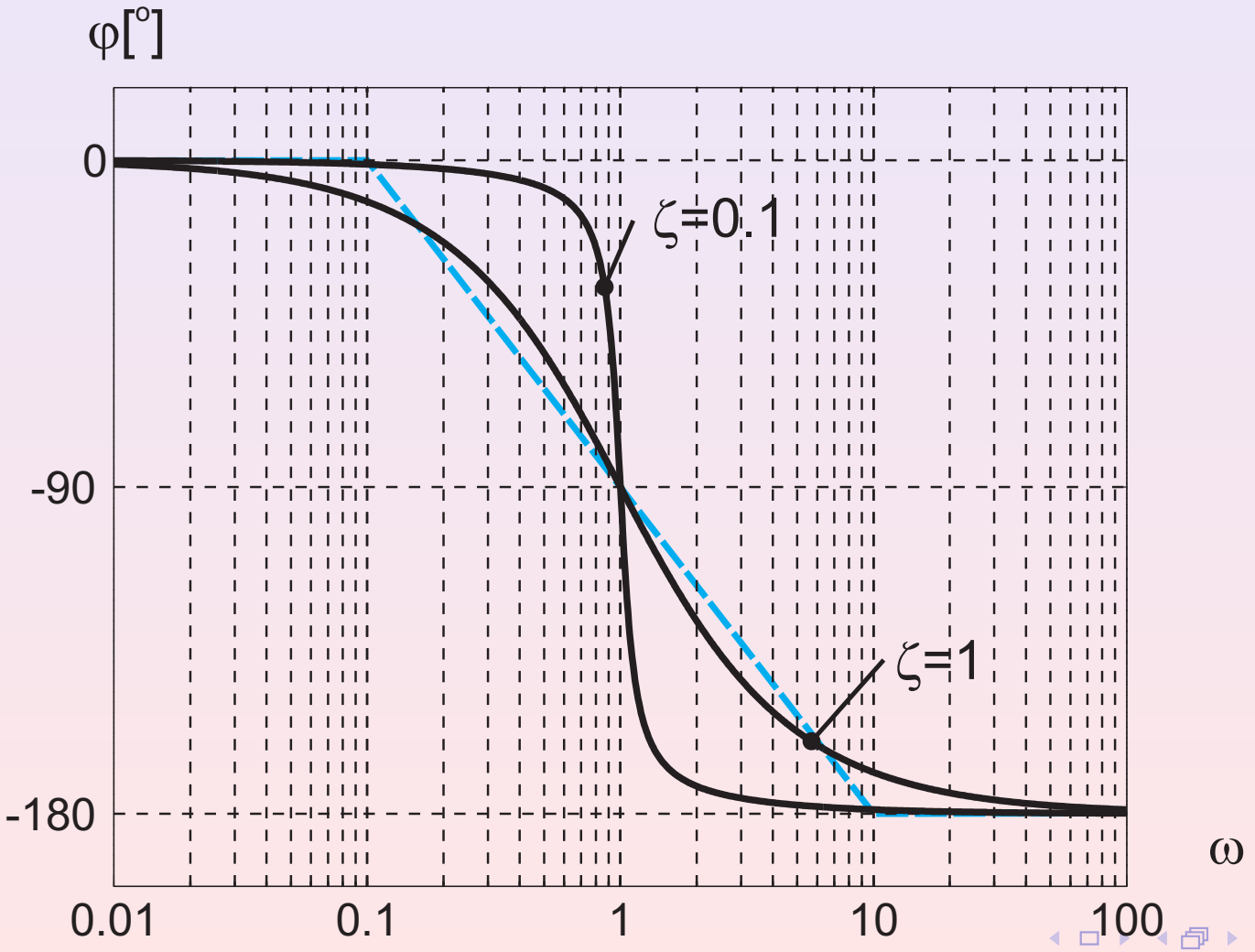


Grafi

20 log |P(jω)| [dB]



Grafi



Splošni Bodejev diagram 2

Izhodišče: $P(j\omega) = |P(j\omega)|e^{j\varphi} = \prod_{i=1}^{n} P_i(j\omega) = \prod_{i=1}^{n} |P_i(j\omega)|e^{j\varphi_i}$

Velja: $|P(j\omega)| = \prod_{i=1}^{n} |P_i(j\omega)|$

in $\varphi = \sum_{i=1}^{n} \varphi_i$

V ampl. frekv. diag. vnašamo $20 \log |P(j\omega)|$, kar nam da:

$$20 \log |P(j\omega)| = 20 \log \left(\prod_{i=1}^{n} |P_i(j\omega)| \right) = \sum_{i=1}^{n} 20 \log |P_i(j\omega)|$$

Splošni Bodejev diagram - Primer

Nariši asimptotski Bodejev diagram za sistem s frekvenčno karakteristiko:

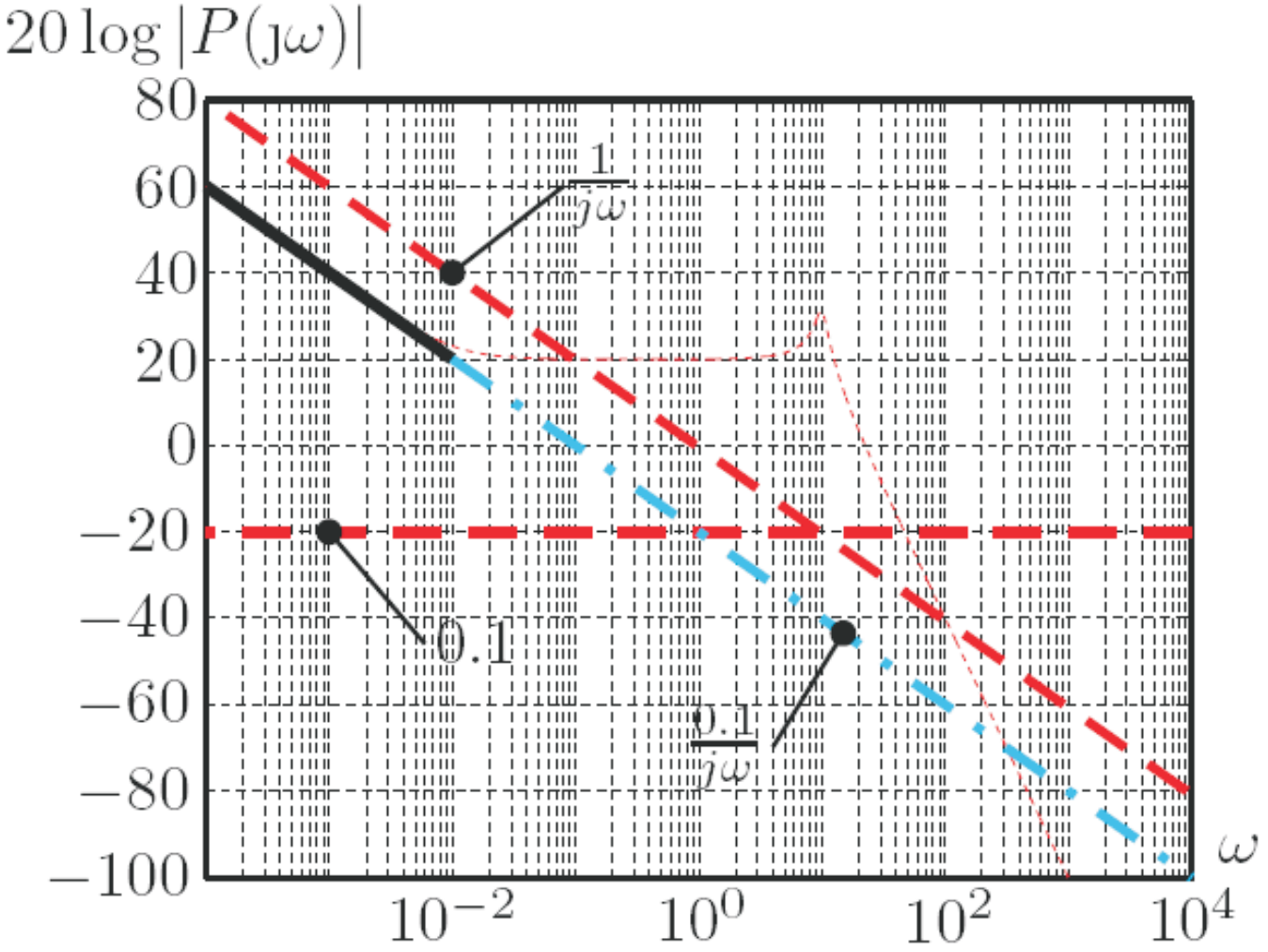
$$P(j\omega) = \frac{100\,(j\omega + 0.01)}{j\omega\,(j\omega + 10)\,(0.01(j\omega)^2 + 0.02j\omega + 1)}$$

Dobljene diagrame primerjaj z dejanskimi!

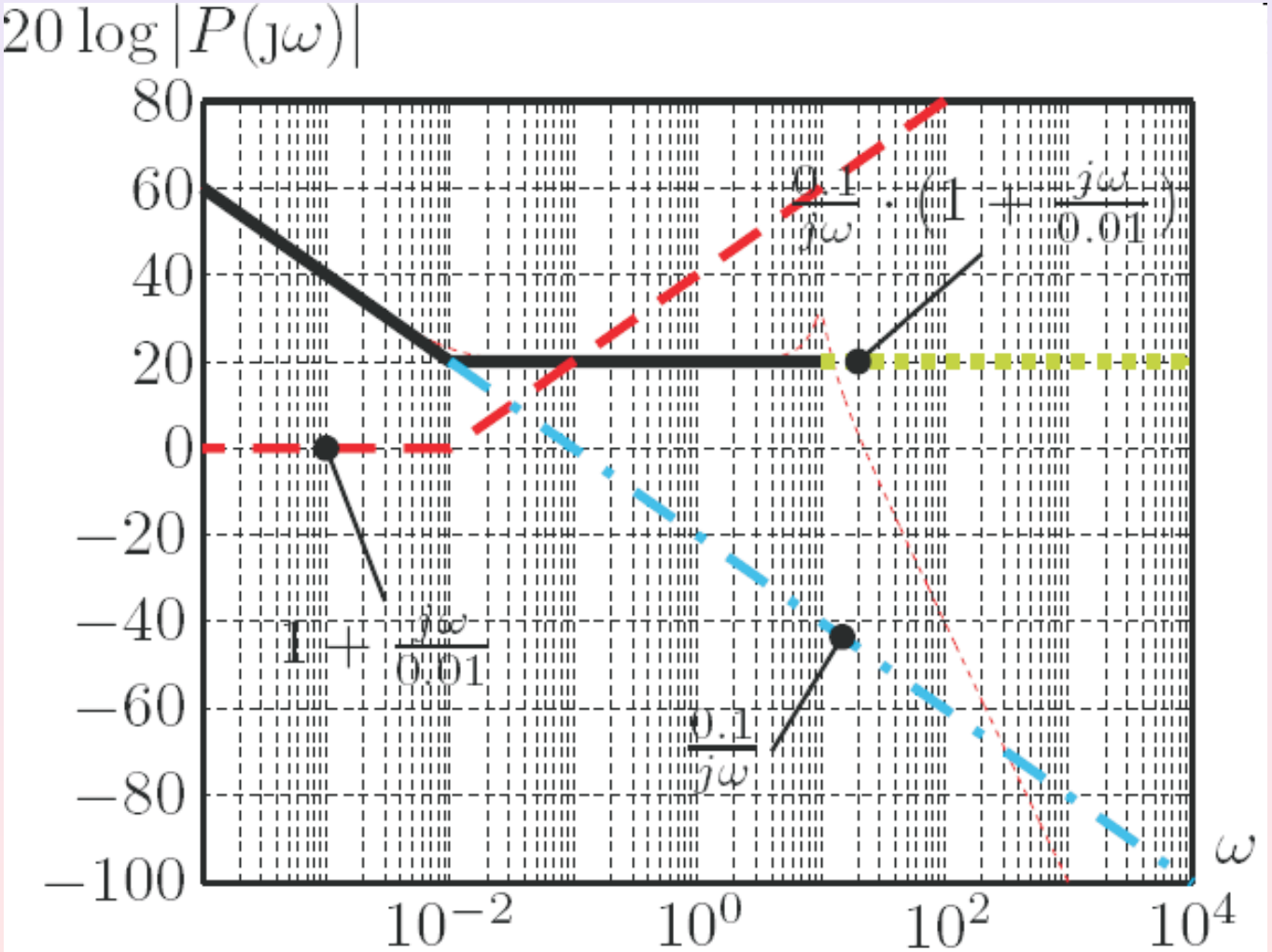
Splošni Bodejev diagram - Primer (tabela)

Faktor	0.1	$1+\frac{j\omega}{0.01}$	$\frac{1}{j\omega}$	$\frac{1}{1+\frac{j\omega}{10}}$	$\frac{1}{1+0.02j\omega+0.01(j\omega)^2}$	Σ
ω_L	\	0.01	\	10	10	\
As. $(\omega \to 0)$	0	0	-20	0	0	-20
As. $(\omega \to \infty)$	0	20	-20	-20	-40	-60
Faza $(\omega \to 0)$	0	0	-90	0	0	-90
Faza $(\omega \to \infty)$	0	90	-90	-90	-180	-270

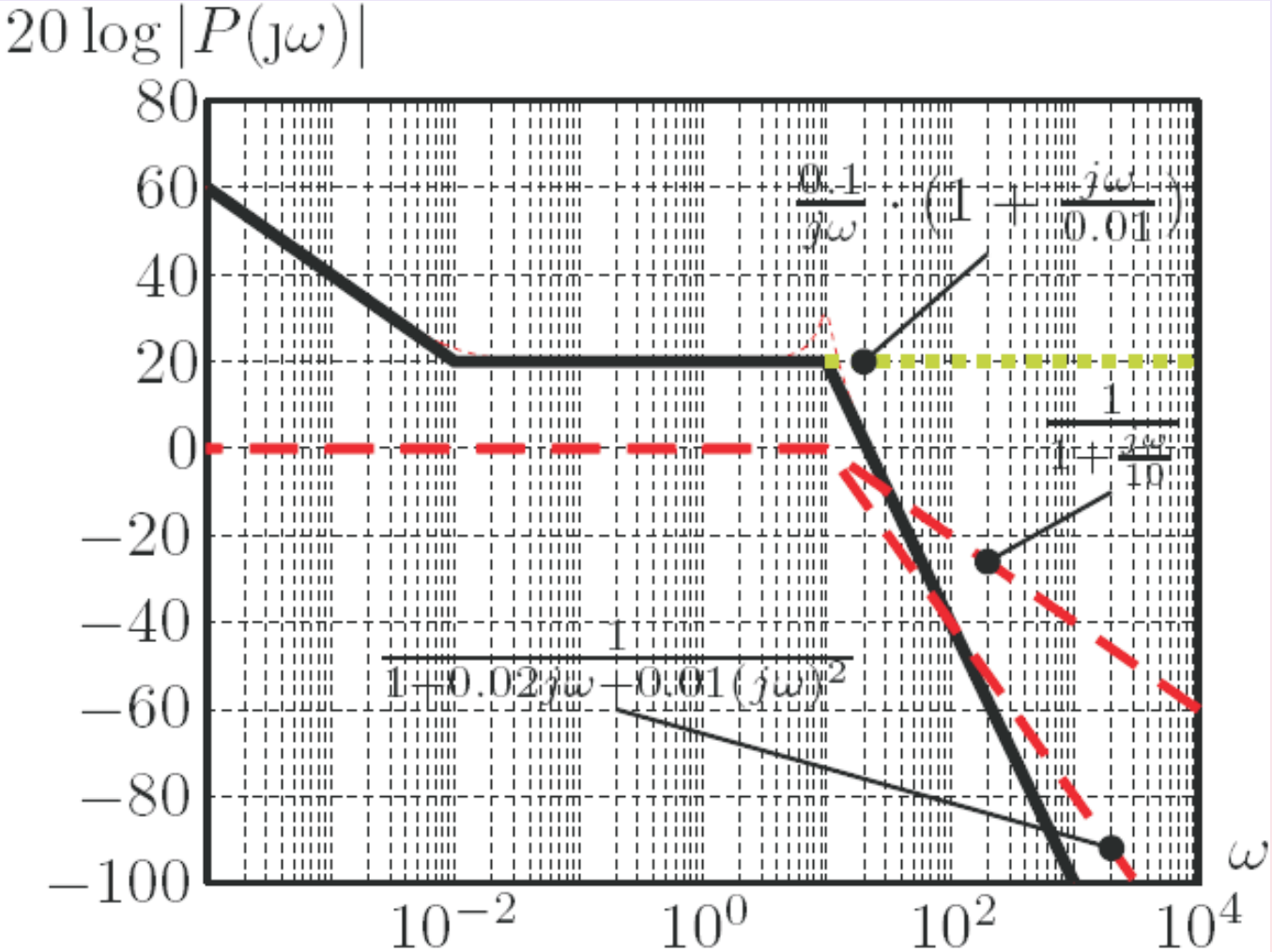
Splošni Bodejev diagram - Primer (konstrukcija)



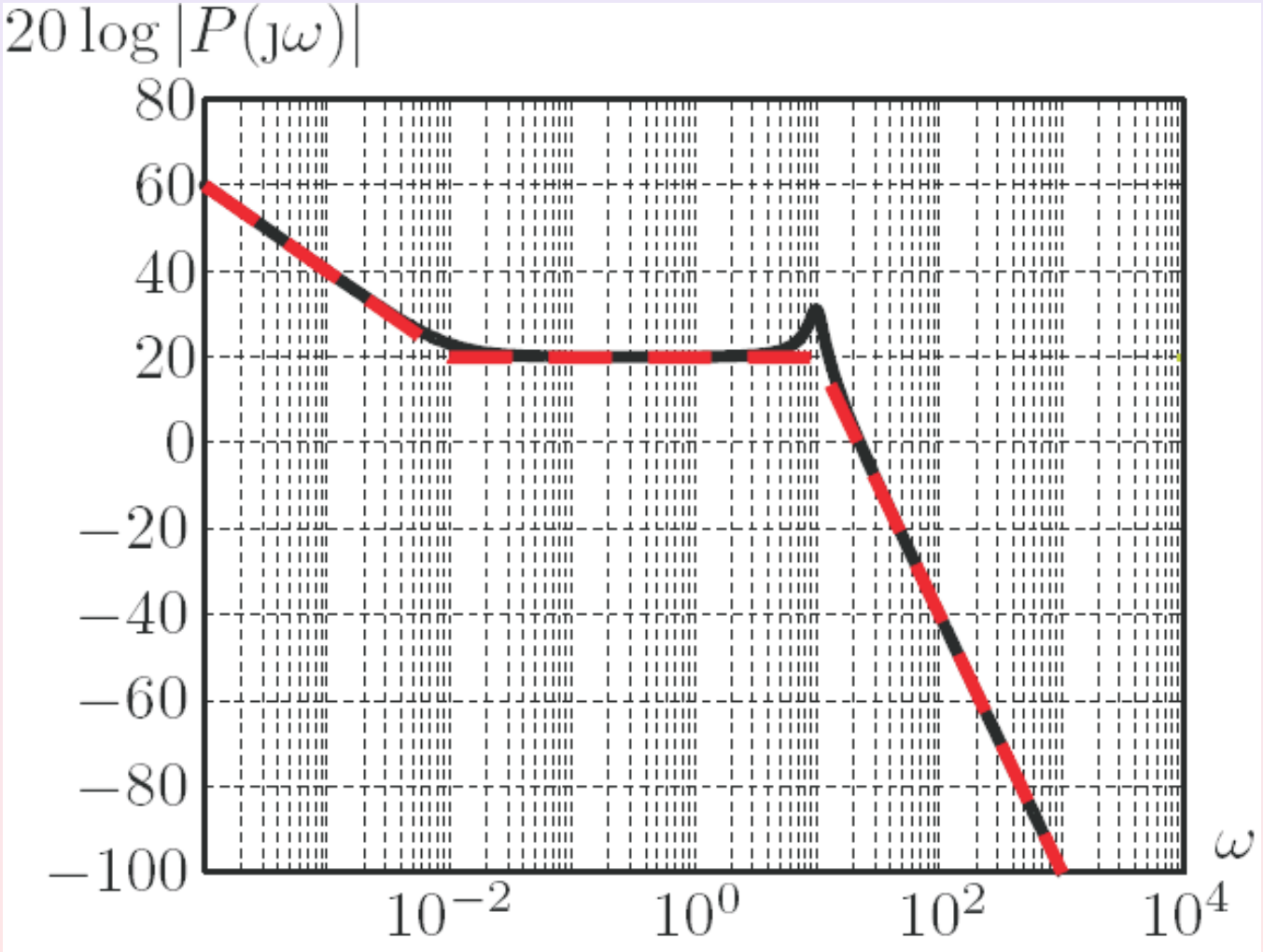
Splošni Bodejev diagram - Primer (konstrukcija)



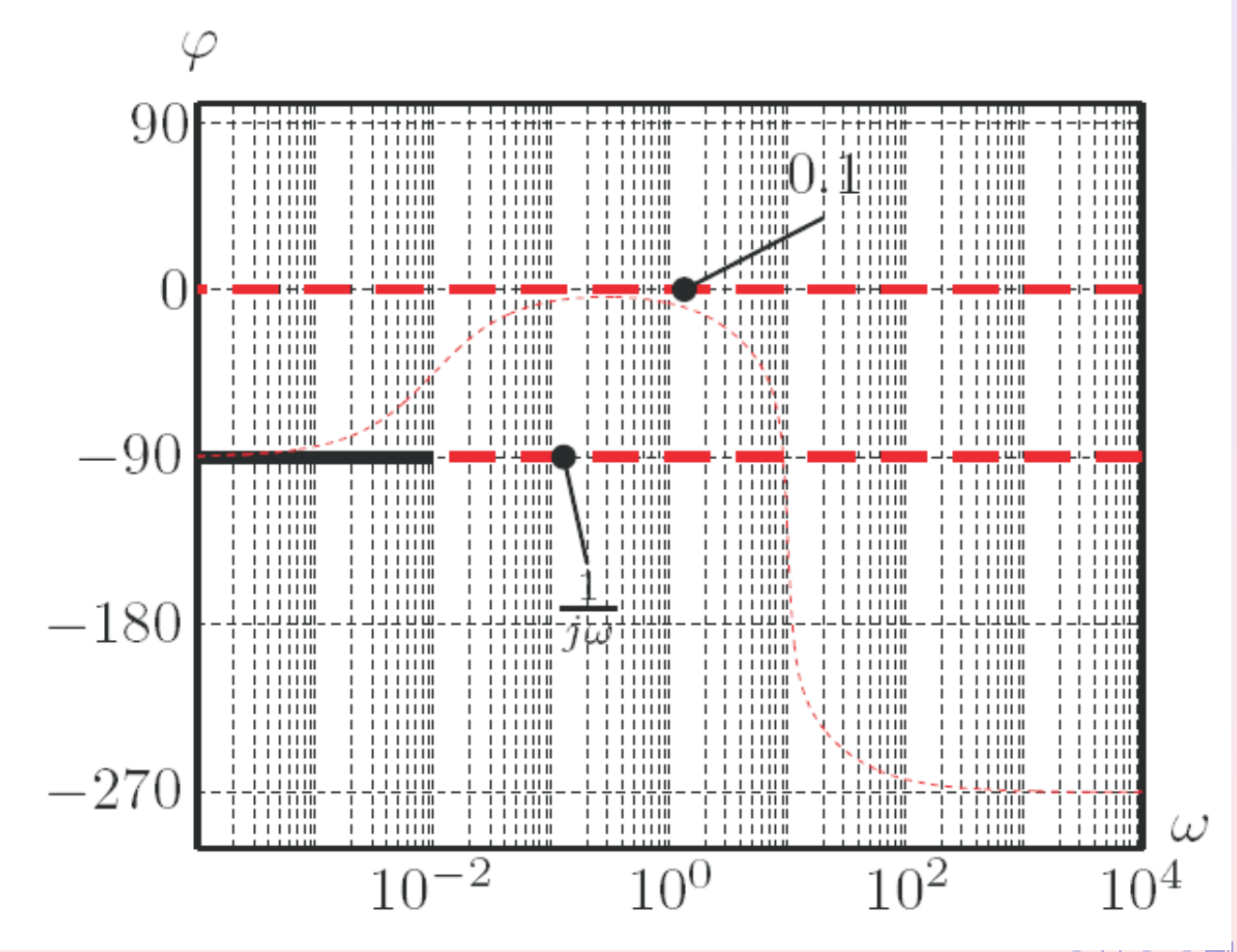
Splošni Bodejev diagram - Primer (konstrukcija)



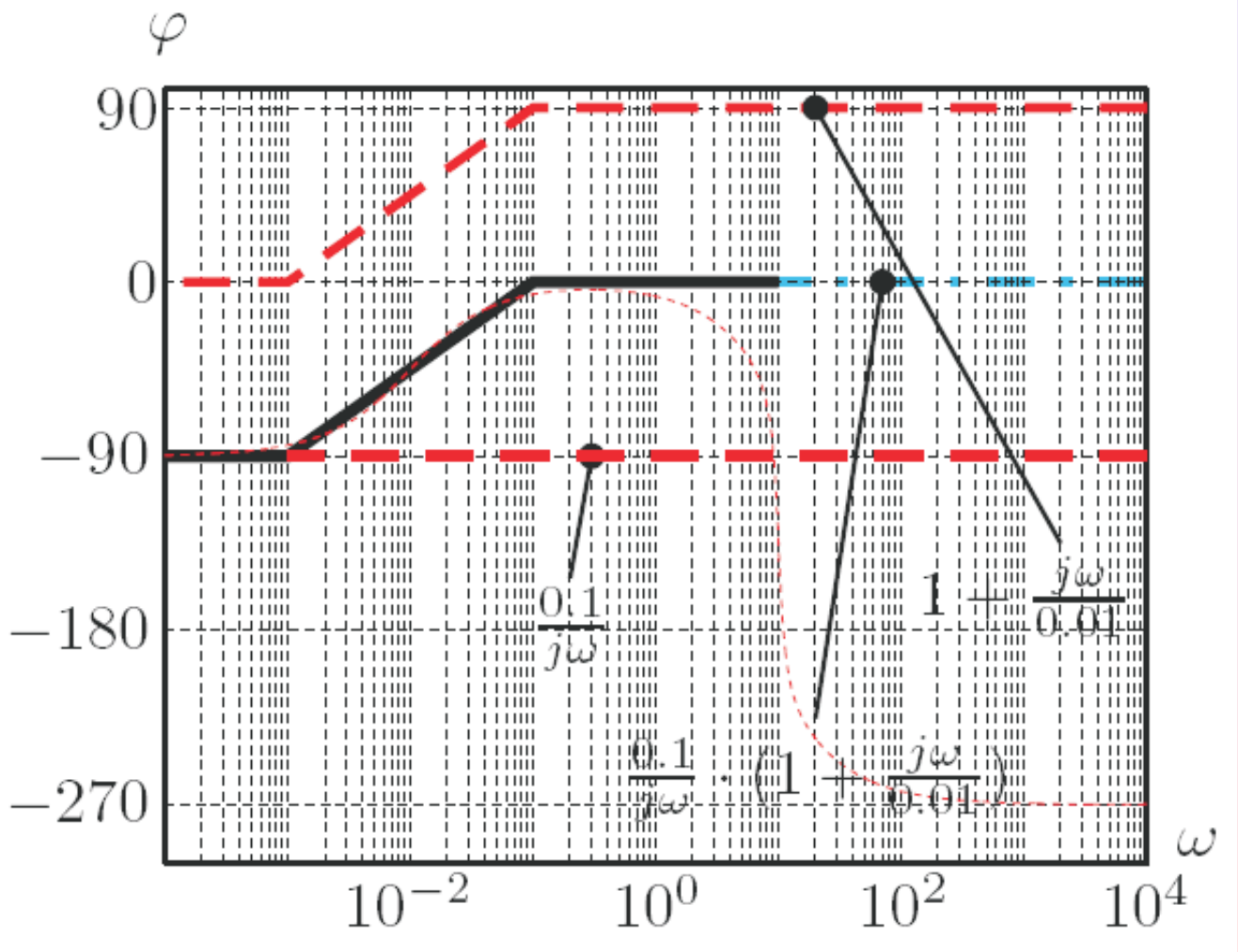
Splošni Bodejev diagram - Primer (konstrukcija)



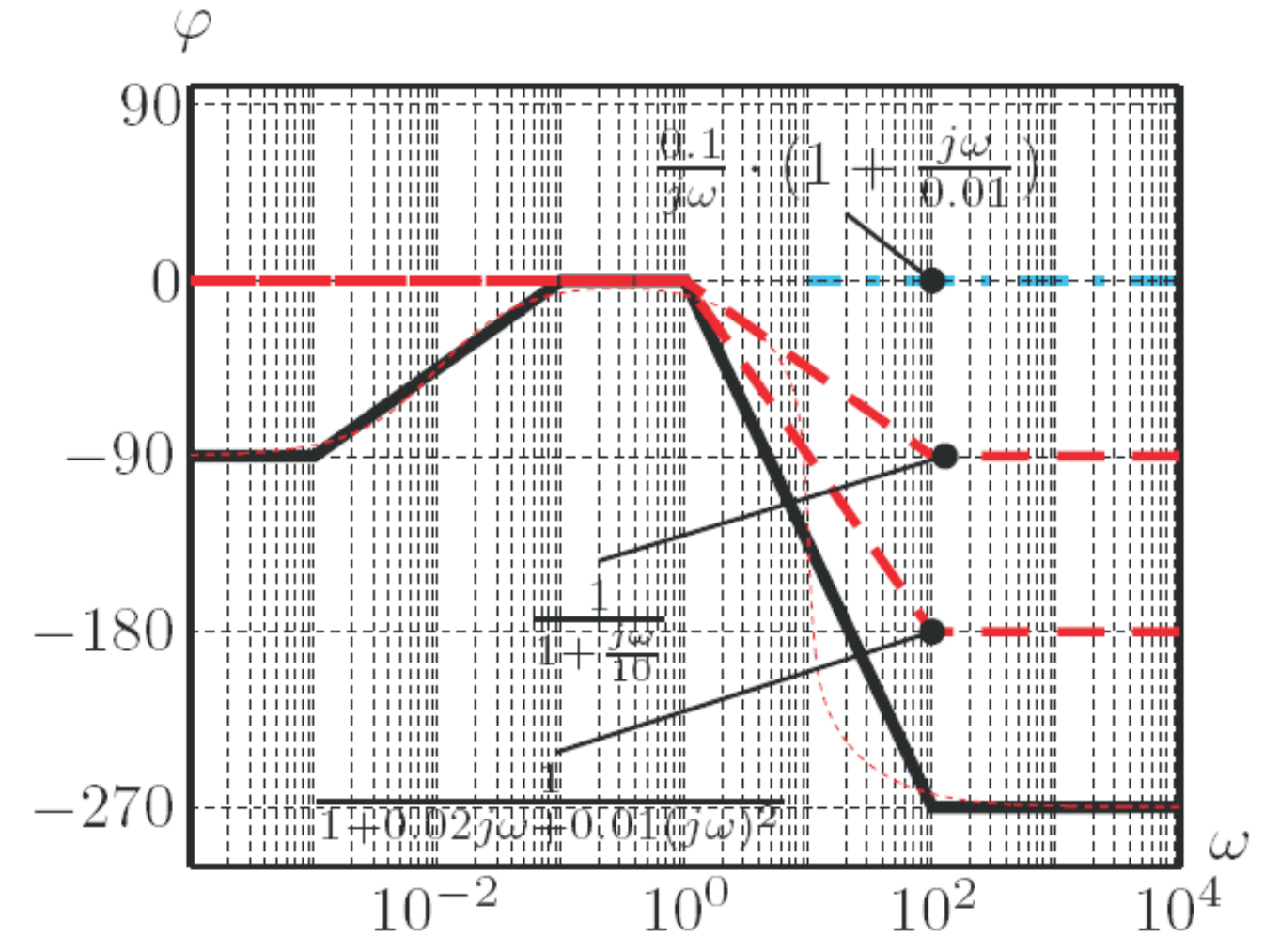
Splošni Bodejev diagram - Primer (konstrukcija)



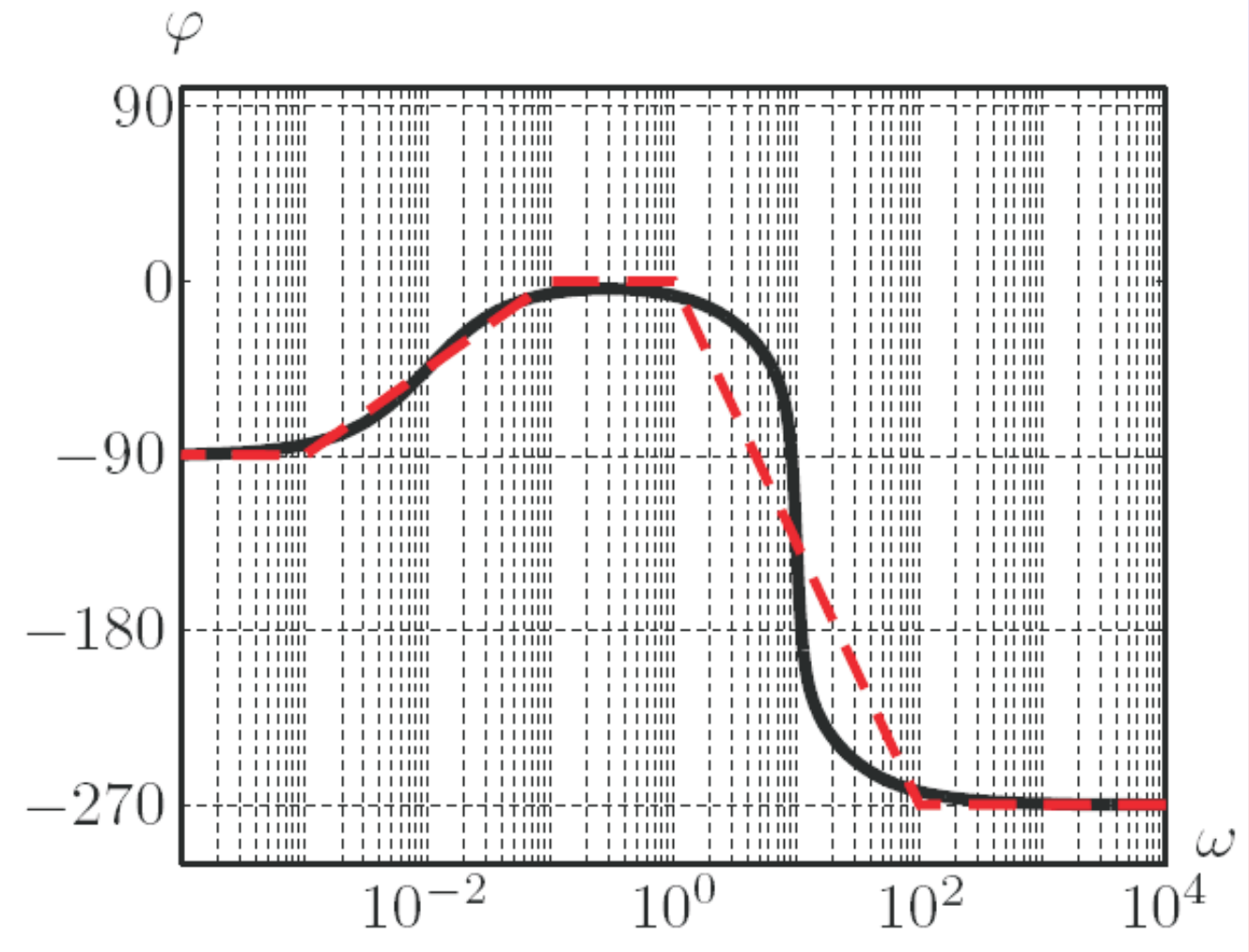
Splošni Bodejev diagram - Primer (konstrukcija)



Splošni Bodejev diagram - Primer (konstrukcija)



Splošni Bodejev diagram - Primer (konstrukcija)



Nyquistov diagram 1

Nyquistov ali polarni diagram je frekvenčni diagram, ki ga rišemo v kompleksni (Gaussovi) ravnini. V tem primeru upoštevamo, da lahko frekvenčno karakteristiko zapišemo v obliki

$$P(j\omega) = |P(j\omega)|e^{j\angle P(j\omega)} = |P(j\omega)|e^{j\varphi(\omega)}$$

Točka v Gaussovi ravnini je del Nyquistovega diagrama, če obstaja frekvenca ω_0, za katero je oddaljenost točke od izhodišča enaka $|P(j\omega_0)|$, njen argument pa $\angle P(j\omega_0)$.

Nyquistov diagram 2

Kot izhodišče za risanje Nyquistovega diagrama se ponavadi vzame frekvenčna karakteristika zapisani v naslednji obliki

$$P(j\omega) = \frac{b_m(j\omega)^m + b_{m-1}(j\omega)^{m-1} + \cdots + b_1(j\omega) + b_0}{(j\omega)^r \left(a_{n-r}(j\omega)^{n-r} + a_{n-r-1}(j\omega)^{n-r-1} + \cdots + a_1(j\omega) + a_0\right)}$$

Iz tega zapisa lahko sklepamo, pod kakšnim kotom se graf začne in pod kakšnim kotom se graf konča.

Nyquistov diagram 3 - Aproksimacija pri nizkih frekvencah

Pri zelo nizkih frekvencah ($\omega \to 0$) lahko frekvenčno karakteristiko $P(j\omega)$ aproksimiramo z izrazom

$$P(j\omega) \approx \frac{b_0}{(j\omega)^r \, a_0}$$

kar nam da

$$\lim_{\omega \to 0} \angle P(j\omega) = -r\frac{\pi}{2}$$

Nyquistov diagram 4 - Aproksimacija pri visokih frekvencah

Pri zelo visokih frekvencah ($\omega \to \infty$) lahko frekvenčno karakteristiko $P(j\omega)$ aproksimiramo z izrazom

$$P(j\omega) \approx \frac{b_m(j\omega)^m}{a_{n-r}(j\omega)^n}$$

kar nam da

$$\lim_{\omega \to \infty} \angle P(j\omega) = -(n-m)\frac{\pi}{2}$$

Ker za realne sisteme velja $n \geq m$, lahko sklepamo, da se graf pri visokih frekvencah zaključi v izhodišču.

Nyquistov diagram 5

V primerjavi z Bodejevim diagramom je pri Nyquistovem diagramu potrebno izpostaviti dve bistveni razliki.

- Nyquistov diagram za razliko od Bodejevega nima amplitudno frekvenčnega in fazno frekvenčnega dela, kar pomeni, da moramo na enem dvodimenzionalnem grafu prikazati odvisnost med tremi spremenljivkami.
- Pri Nyquistovem diagramu se ne uporabljajo asimptotski diagrami, kar pomeni, da ga za bolj komplicirane frekvenčne karakteristike rišemo z računalnikom.

Za bolj nazorno predstavo si poglejmo dva primera.

Nyquistov diagram - Primer 1 - 1

Nariši Nyquistov diagram za sistem s frekvenčno karakteristiko

$$P(j\omega) = \frac{1}{1 + 0.1\, j\omega}$$

Preveri rezultat z aproksimacijami pri visokih in nizkih frekvencah.

Nyquistov diagram - Primer 1 - 2

Iz frekvenčne karakteristike najprej določimo realno in imaginarno komponento

$$P(j\omega) = \frac{1}{1+0.1\,j\omega} = \frac{1-0.1\,j\omega}{1+0.01\omega^2} \Rightarrow \begin{cases} \mathrm{Re} = \dfrac{1}{1+0.01\omega^2} \\ \mathrm{Im} = \dfrac{-0.1\omega}{1+0.01\omega^2} \end{cases}$$

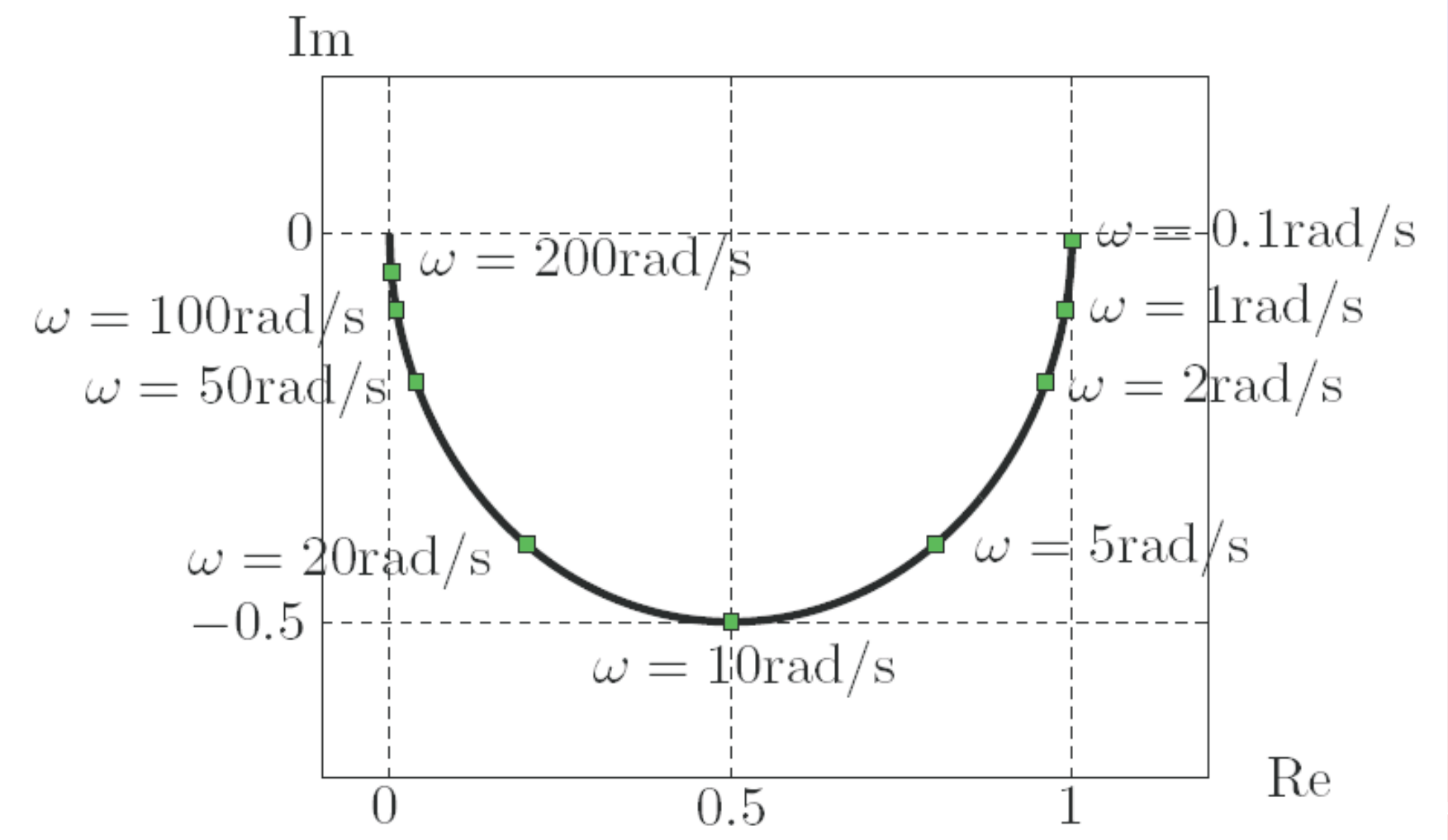
Nato obe komponenti za nekaj frekvenc izračunamo (glej tabelo).

Nyquistov diagram - Primer 1 - 3

ω	0.1	1	2	5	10	20	50	100	200
Re	1.00	0.99	0.96	0.80	0.50	0.20	0.04	0.01	0.00
Im	-0.01	-0.10	-0.19	-0.40	-0.50	-0.40	-0.19	-0.10	-0.05

.

Nyquistov diagram - Primer 1 - 4



Nyquistov diagram - Primer 1 - 5

Če prenosno karakteristiko $P(j\omega) = \frac{1}{1+0.1\,j\omega}$ primerjamo z izrazom

$$P(j\omega) = \frac{b_m(j\omega)^m + b_{m-1}(j\omega)^{m-1} + \cdots + b_1(j\omega) + b_0}{(j\omega)^r\ (a_{n-r}(j\omega)^{n-r} + a_{n-r-1}(j\omega)^{n-r-1} + \cdots + a_1(j\omega) + a_0)}$$

lahko ugotovimo, da velja $n = 1$, $r = 0$ in $m = 0$. Pri nizkih frekvencah torej velja

$$\lim_{\omega\to 0} \angle P(j\omega) = -r\frac{\pi}{2} = 0$$

Graf se torej začne pod kotom 0°. Pri visokih frekvencah velja

$$\lim_{\omega\to\infty} \angle P(j\omega) = -(n-m)\frac{\pi}{2} = -(1-0)\frac{\pi}{2} = -\frac{\pi}{2}$$

Ker je stopnja imenovalca večja od stopnje števca ($n > m$), se graf pri visokih frekvencah približuje izhodišču in sicer pod kotom -90°. V danem primeru ima graf obliko polkroga.

Nyquistov diagram - Primer 2 - 1

Nariši Nyquistov diagram za sistem s frekvenčno karakteristiko

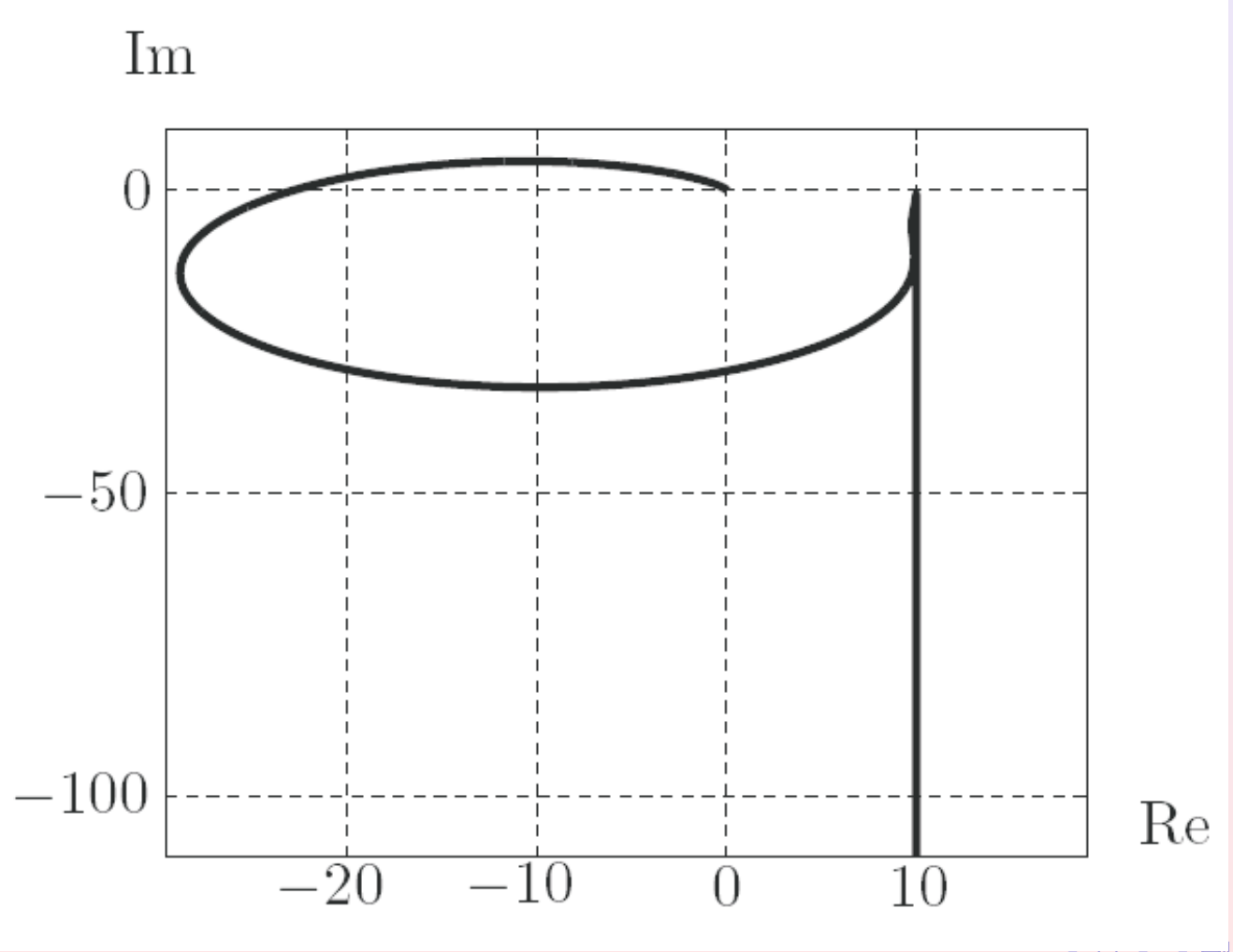
$$P(j\omega) = \frac{100\,(j\omega + 0.01)}{j\omega\,(j\omega + 10)\,(0.01(j\omega)^2 + 0.02j\omega + 1)}$$

Preveri rezultat z aproksimacijami pri visokih in nizkih frekvencah.

Nyquistov diagram - Primer 2 - 2

```
clear
w=logspace(-4,4,100000);
s=i*w;
P=100*(s+0.01)./(s.*(s+10).*(0.01*s.^2+0.02*s+1));
amp=abs(P);
faza=unwrap(angle(P));
Re=amp.*cos(faza);
Im=amp.*sin(faza);
plot(Re,Im,'Color','black','LineWidth',4);
```

Nyquistov diagram - Primer 2 - 3



Nyquistov diagram - Primer 2 - 4

Če prenosno karakteristiko $P(j\omega) = \frac{100\,(j\omega+0.01)}{j\omega\,(j\omega+10)\,(0.01(j\omega)^2+0.02j\omega+1)}$ primerjamo z izrazom

$$P(j\omega) = \frac{b_m(j\omega)^m + b_{m-1}(j\omega)^{m-1} + \cdots + b_1(j\omega) + b_0}{(j\omega)^r\ (a_{n-r}(j\omega)^{n-r} + a_{n-r-1}(j\omega)^{n-r-1} + \cdots + a_1(j\omega) + a_0)}$$

lahko ugotovimo, da velja $n = 4$, $r = 1$ in $m = 1$. Pri nizkih frekvencah torej velja

$$\lim_{\omega\to 0} \angle P(j\omega) = -r\frac{\pi}{2} = -\frac{\pi}{2}$$

Graf se torej začne pod kotom $-90°$. Pri visokih frekvencah velja

$$\lim_{\omega\to\infty} \angle P(j\omega) = -(n-m)\frac{\pi}{2} = -(4-1)\frac{\pi}{2} = -3\frac{\pi}{2}$$

Ker je stopnja imenovalca večja od stopnje števca ($n > m$), se graf pri visokih frekv. približuje izhodišču in sicer pod kotom $-270°$.